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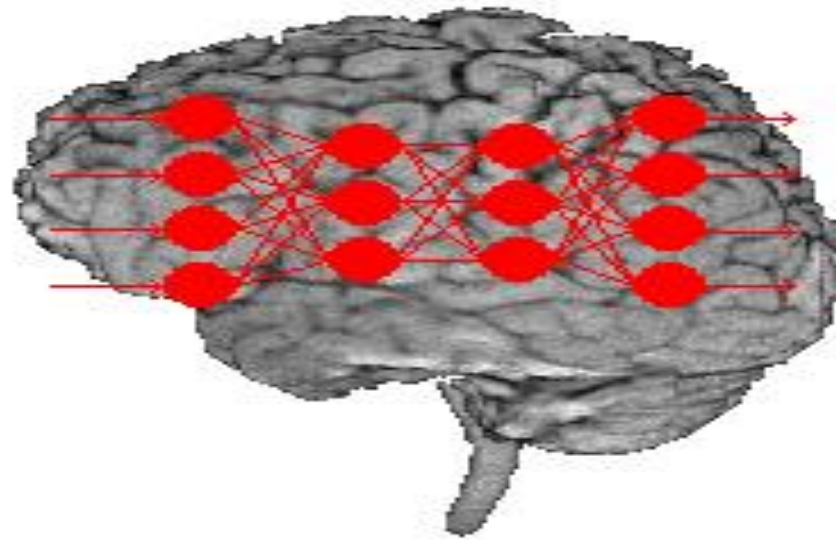
ARTIFICIAL NEURAL NETWORK AND FUZZY LOGIC

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Artificial Neural Networks



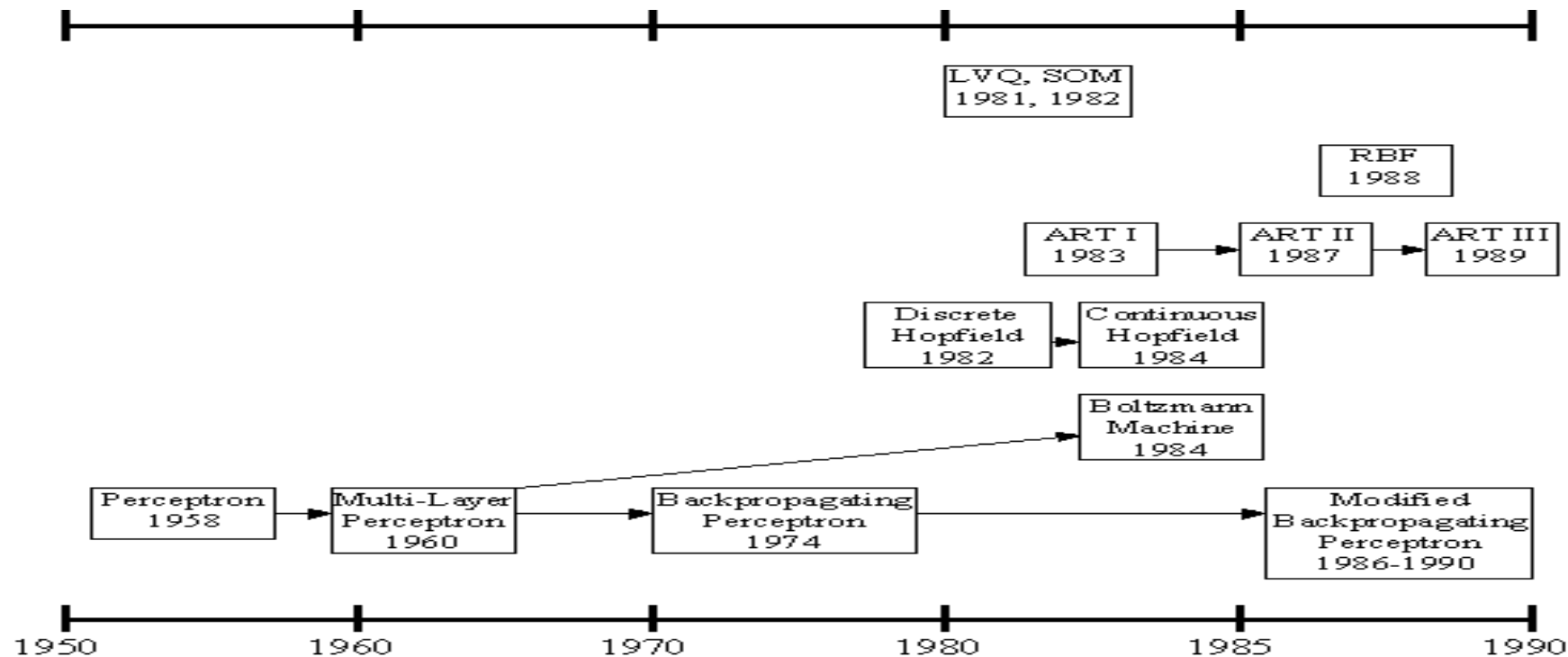
Ashwini Sawant

Agenda

- History of Artificial Neural Networks
- What is an Artificial Neural Networks?
- How it works?
- Learning
 - Learning paradigms
 - Supervised learning
 - Unsupervised learning
 - Reinforcement learning
 - Applications areas
 - Advantages and Disadvantages

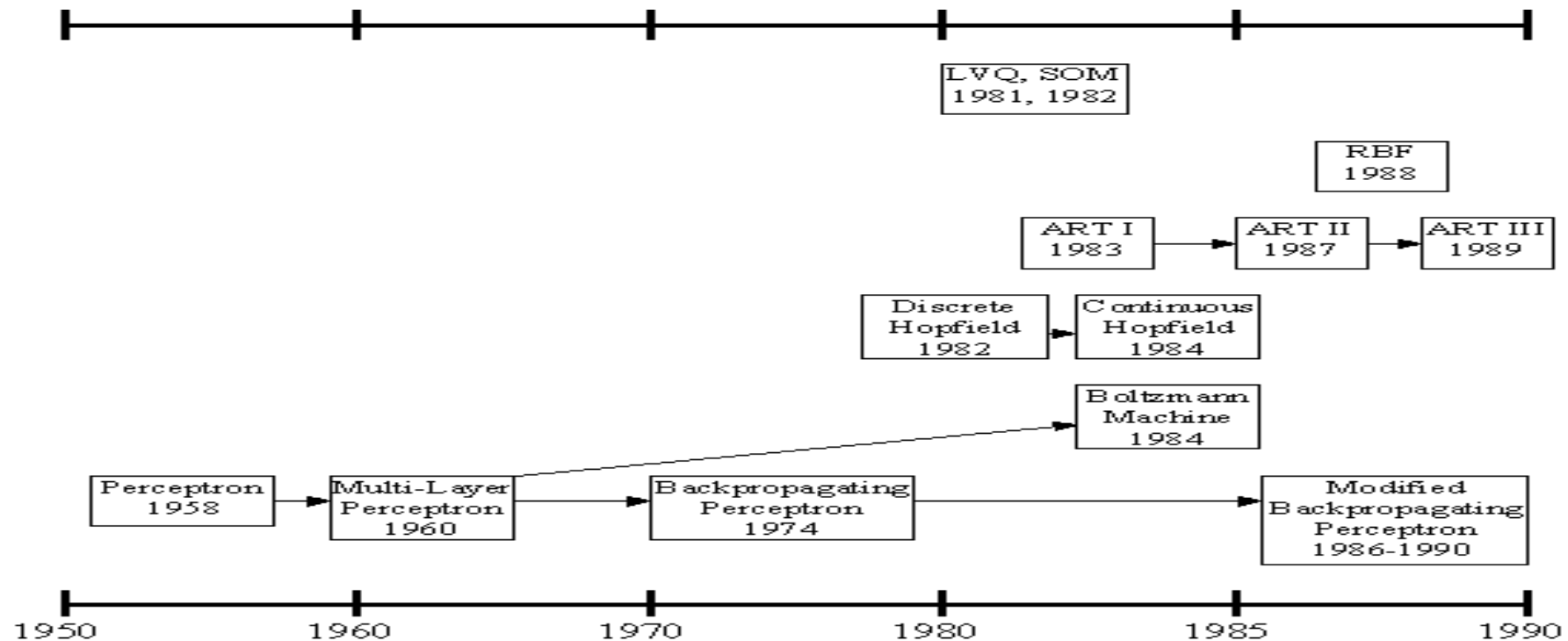
History of the Artificial Neural Networks

- History of the ANNs stems from the 1940s, the decade of the first electronic computer.
- However, the first important step took place in 1957 when Rosenblatt introduced the first concrete neural model, the perceptron. Rosenblatt also took part in constructing the first successful neurocomputer, the Mark I Perceptron. After this, the development of ANNs has proceeded as described in *Figure*.



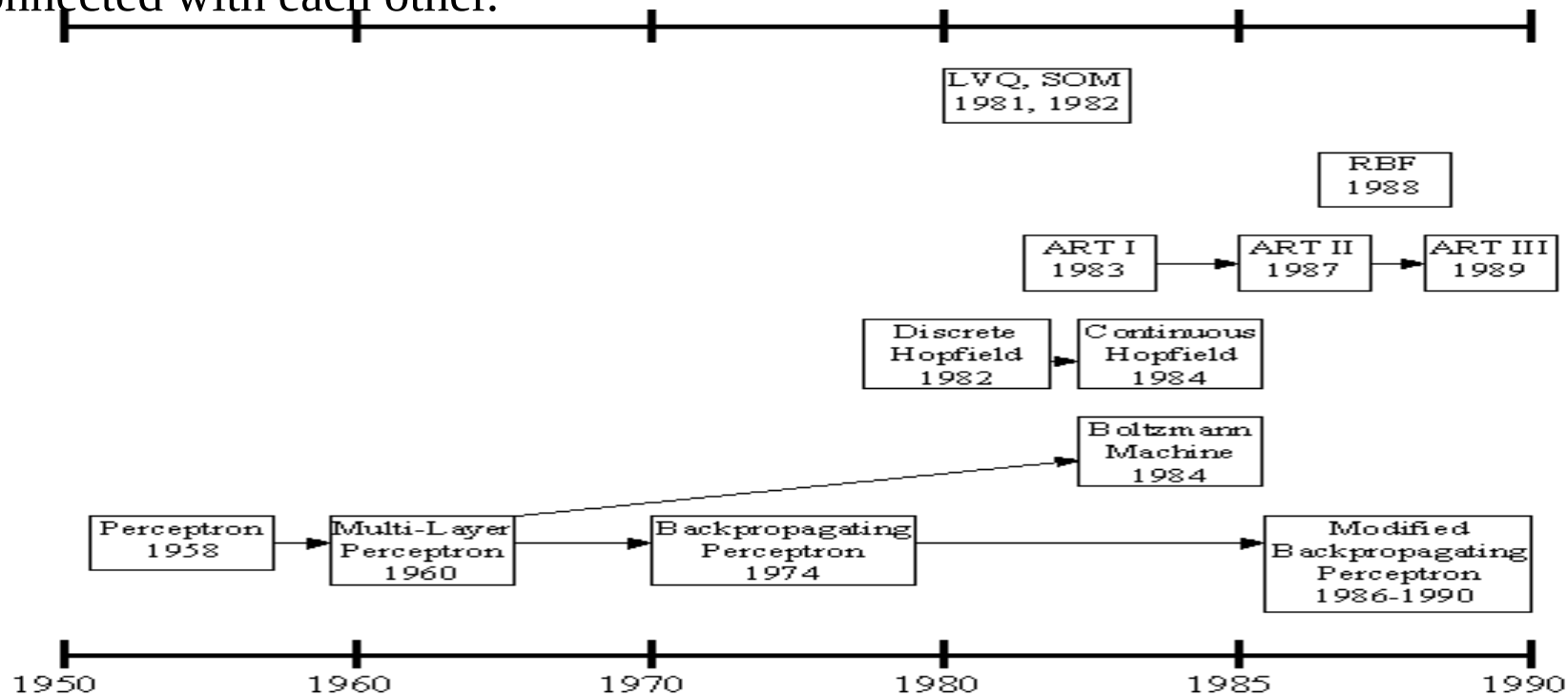
History of the Artificial Neural Networks

- Rosenblatt's original perceptron model contained only one layer. From this, a multi-layered model was derived in 1960. At first, the use of the multi-layer perceptron (MLP) was complicated by the lack of an appropriate learning algorithm.
- In 1974, Werbos came to introduce a so-called backpropagation algorithm for the three-layered perceptron network.



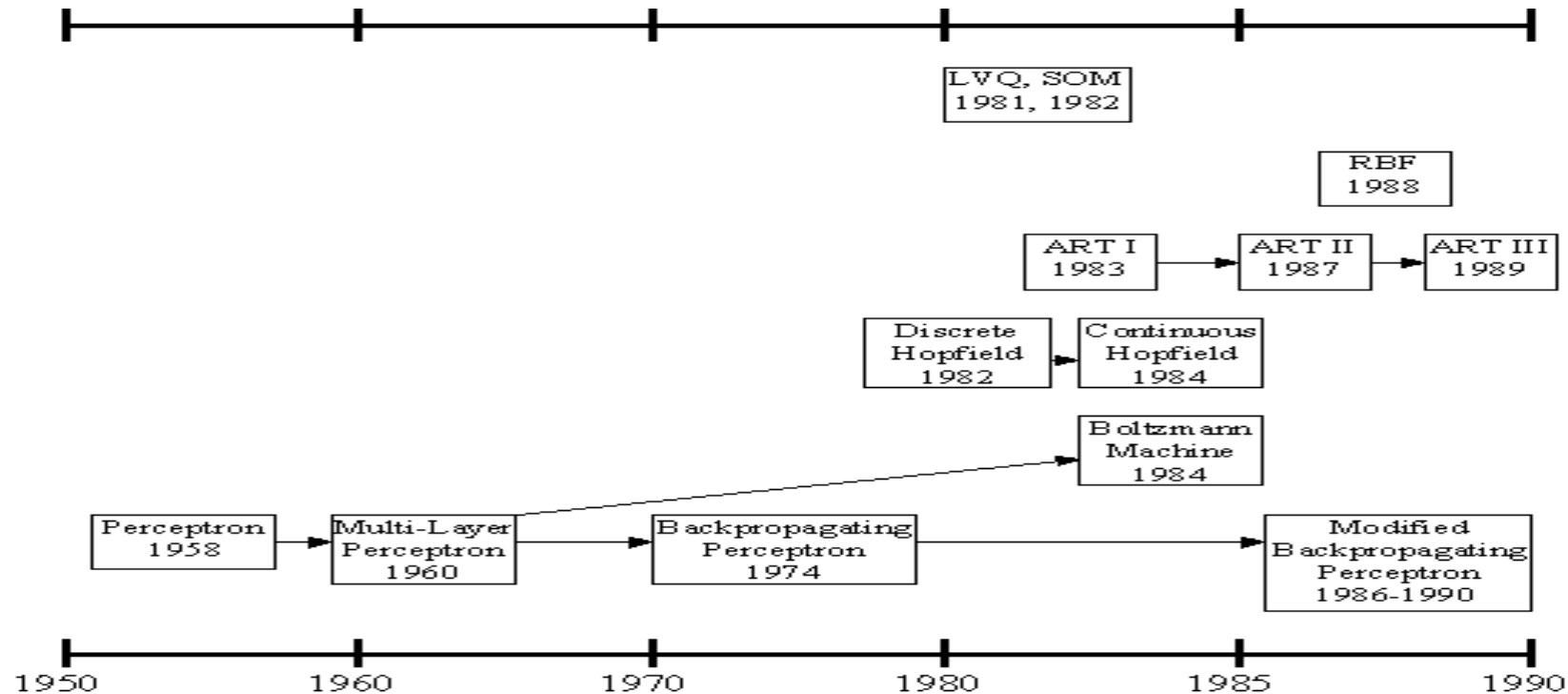
History of the Artificial Neural Networks

- In 1986, The application area of the MLP networks remained rather limited until the breakthrough when a general back propagation algorithm for a multi-layered perceptron was introduced by Rummelhart and Mclelland.
- in 1982, Hopfield brought out his idea of a neural network. Unlike the neurons in MLP, the Hopfield network consists of only one layer whose neurons are fully connected with each other.



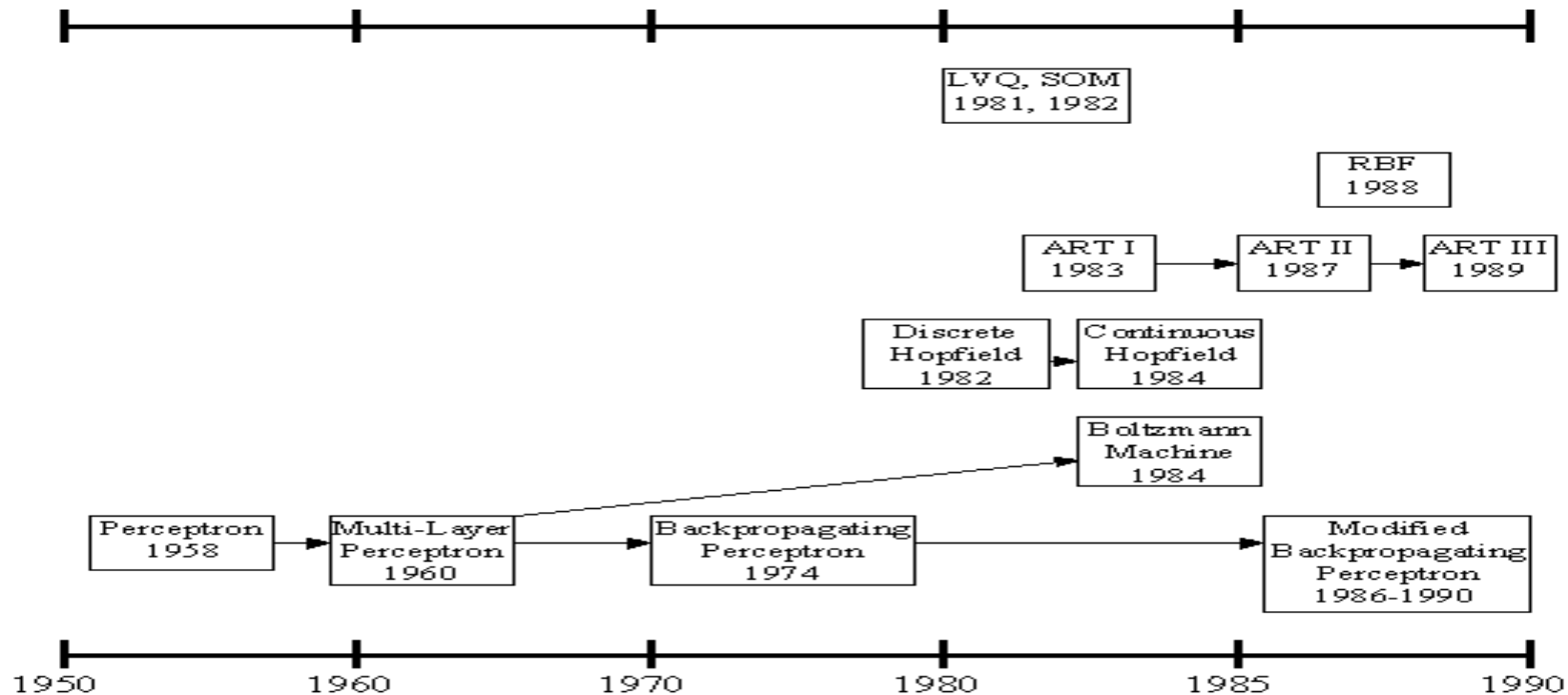
History of the Artificial Neural Networks

- Since then, new versions of the Hopfield network have been developed. The Boltzmann machine has been influenced by both the Hopfield network and the MLP.



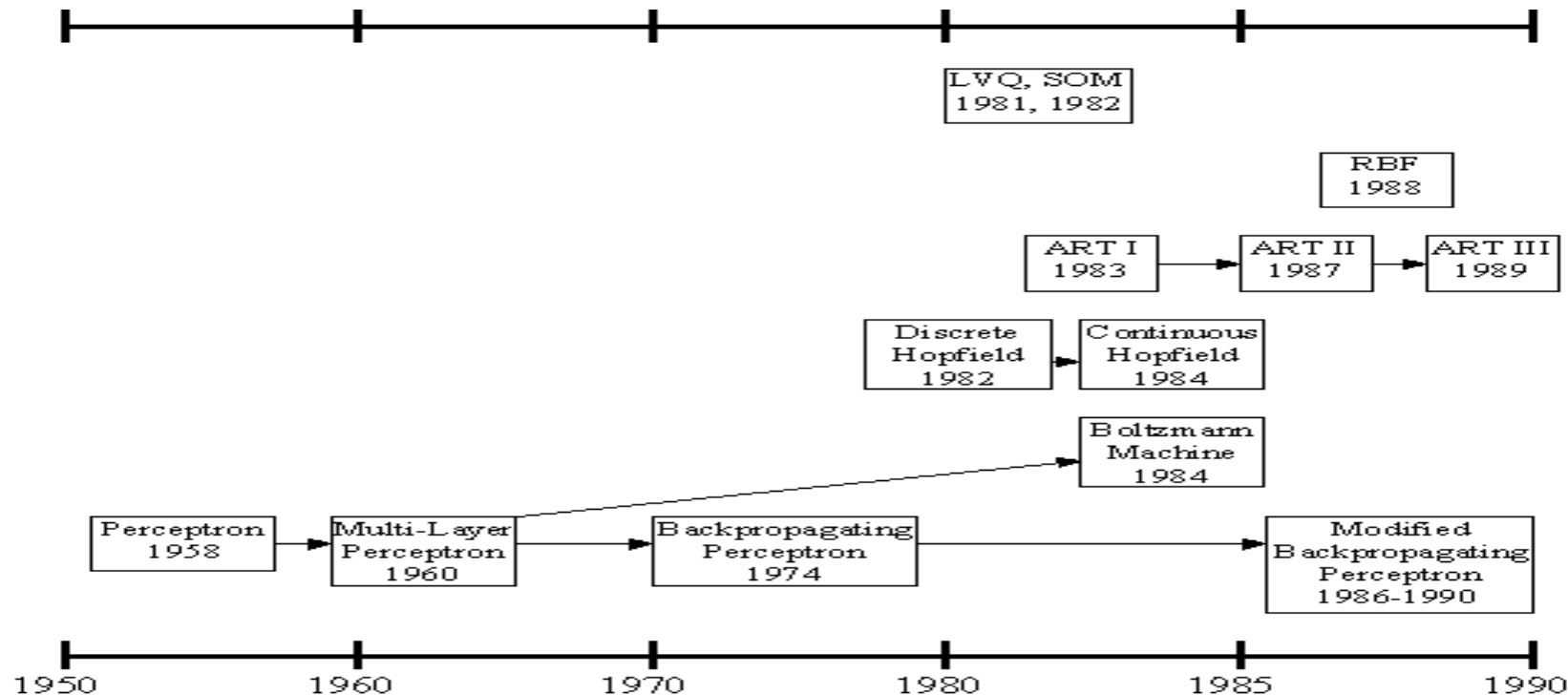
History of the Artificial Neural Networks

- in 1988, Radial Basis Function (RBF) networks were first introduced by Broomhead & Lowe. Although the basic idea of RBF was developed 30 years ago under the name method of potential function, the work by Broomhead & Lowe opened a new frontier in the neural network community.



History of the Artificial Neural Networks

- in 1982, A totally unique kind of network model is the Self-Organizing Map (SOM) introduced by Kohonen. SOM is a certain kind of topological map which organizes itself based on the input patterns that it is trained with. The SOM originated from the LVQ (Learning Vector Quantization) network the underlying idea of which was also Kohonen's in 1972.



History of Artificial Neural Networks

Since then, research on artificial neural networks has remained active, leading to many new network types, as well as hybrid algorithms and hardware for neural information processing.

EVOLUTION OF NEURAL NETWORKS

Year	Neural network	Designer	Description
1943	McCulloch and Pitts neuron	McCulloch and Pitts	Arrangement of neurons is combination of logic gate. Unique feature is thresh hold
1949	Hebb network	Hebb	If two neurons are active, then their connection strengths should be increased.
1958,1959,1962,1988,1960	Perceptron Adaline	Frank Rosenblatt, Block, Minsky and Papert Widrow and Hoff	Weights are adjusted to reduce the difference between the net input to the output unit and the desired output

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1972	Kohonen self-organizing feature map	Kohonen	Inputs are clustered to obtain a fired output neuron.
1982, 1984, 1985, 1986, 1987	Hopfield network	John Hopfield and Tank	Based on fixed weights. Can act as associative memory nets
1986	Back propagation network	Rumelhart, Hinton and Williams	i) Multilayered ii) Error propagated backward from output to the hidden units

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1988	Counter propagation network	Grossberg	Similar to kohonen network
1987-1990	Adaptive resonance Theory(ART)	Carpenter and Grossberg	Designed for binary and analog inputs.
1988	Radial basis function network	Broomhead and Lowe	Resemble back propagation network , but activation function used is Gaussian function
1988	Neo cognitron	Fukushima	For character recognition.

Artificial Neural Network

- An artificial neural network consists of a pool of simple processing units which communicate by sending signals to each other over a large number of weighted connections.
- A Neural Network is a machine that is designed to model the way in which the brain performs a particular task or function of interest; the network is usually implemented by using electronic components or is simulated in software on a digital computer.

Artificial Neural Network

- A set of major aspects of a parallel distributed model include:
 - a set of processing units (cells).
 - a state of activation for every unit, which equivalent to the output of the unit.
 - connections between the units. Generally each connection is defined by a weight.
 - a propagation rule, which determines the effective input of a unit from its external inputs.
 - an activation function, which determines the new level of activation based on the effective input and the current activation.
 - an external input for each unit.
 - a method for information gathering (the learning rule).
 - an environment within which the system must operate, providing input signals and _ if necessary _ error signals.

Computers vs. Neural Networks

“Standard” Computers

- one CPU
- fast processing units
- reliable units
- static infrastructure

Neural Networks

highly parallel processing

slow processing units

unreliable units

dynamic infrastructure

BRAIN VS COMPUTER

Term	Brain	Computer
Speed	Execution time is few milliseconds	Execution time is few nano seconds
Processing	Perform massive parallel operations simultaneously	Perform several parallel operations simultaneously. It is faster than the biological neuron
Size and complexity	Number of Neuron is 10^{11} and number of interconnections is 10^{15} . So complexity of brain is higher than computer	It depends on the chosen application and network designer.
Storage capacity	i) Information is stored in interconnections or in synapse strength. ii) New information is stored without destroying old one. iii) Sometimes fails to recollect information	i) Stored in continuous memory location. ii) Overloading may destroy older locations. iii) Can be easily retrieved

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Tolerance	i) Fault tolerant ii) Store and retrieve information even interconnections fails iii) Accept redundancies	i) No fault tolerance ii) Information corrupted if the network connections disconnected. iii) No redundancies
Control mechanism	Depends on active chemicals and neuron connections are strong or weak	CPU Control mechanism is very simple

Why Artificial Neural Networks?

- There are two basic reasons why we are interested in building artificial neural networks (ANNs):
 - **Technical viewpoint:** Some problems such as character recognition or the prediction of future states of a system require massively parallel and adaptive processing.
 - **Biological viewpoint:** ANNs can be used to replicate and simulate components of the human (or animal) brain, thereby giving us insight into natural information processing.

Artificial Neural Networks

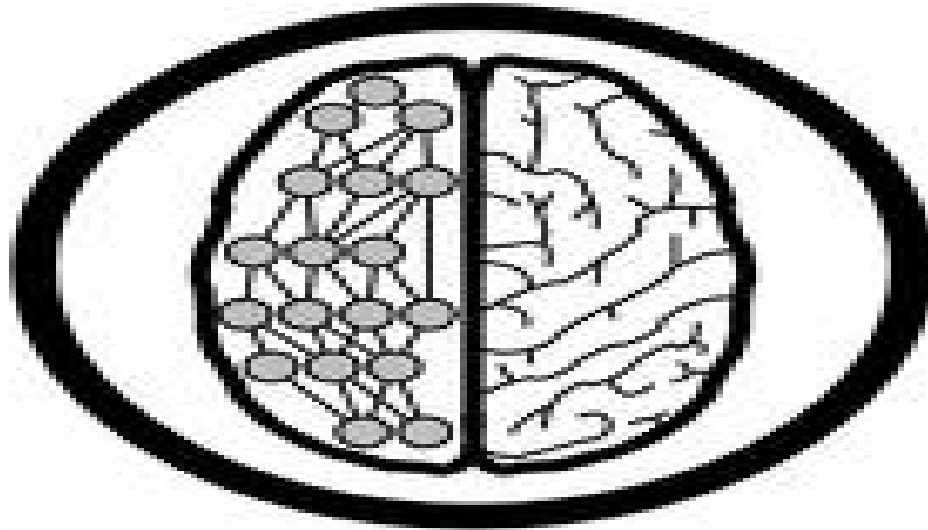
- The “building blocks” of neural networks are the **neurons**.
 - In technical systems, we also refer to them as **units** or **nodes**.
- Basically, each neuron
 - receives **input** from many other neurons.
 - changes its internal state (**activation**) based on the current input.
 - sends **one output signal** to many other neurons, possibly including its input neurons (recurrent network).

Artificial Neural Networks

- Information is transmitted as a series of electric impulses, so-called **spikes**.
- The **frequency** and **phase** of these spikes encodes the information.
- In biological systems, one neuron can be connected to as many as **10,000** other neurons.
- Usually, a neuron receives its information from other neurons in a confined area, its so-called **receptive field**.

How do ANNs work?

- An artificial neural network (ANN) is either a **hardware implementation** or a **computer program** which strives to simulate the information processing capabilities of its biological exemplar. ANNs are typically composed of a great number of interconnected artificial neurons. The artificial neurons are simplified models of their biological counterparts.
- ANN is a technique for solving problems by constructing software that works like our brains.



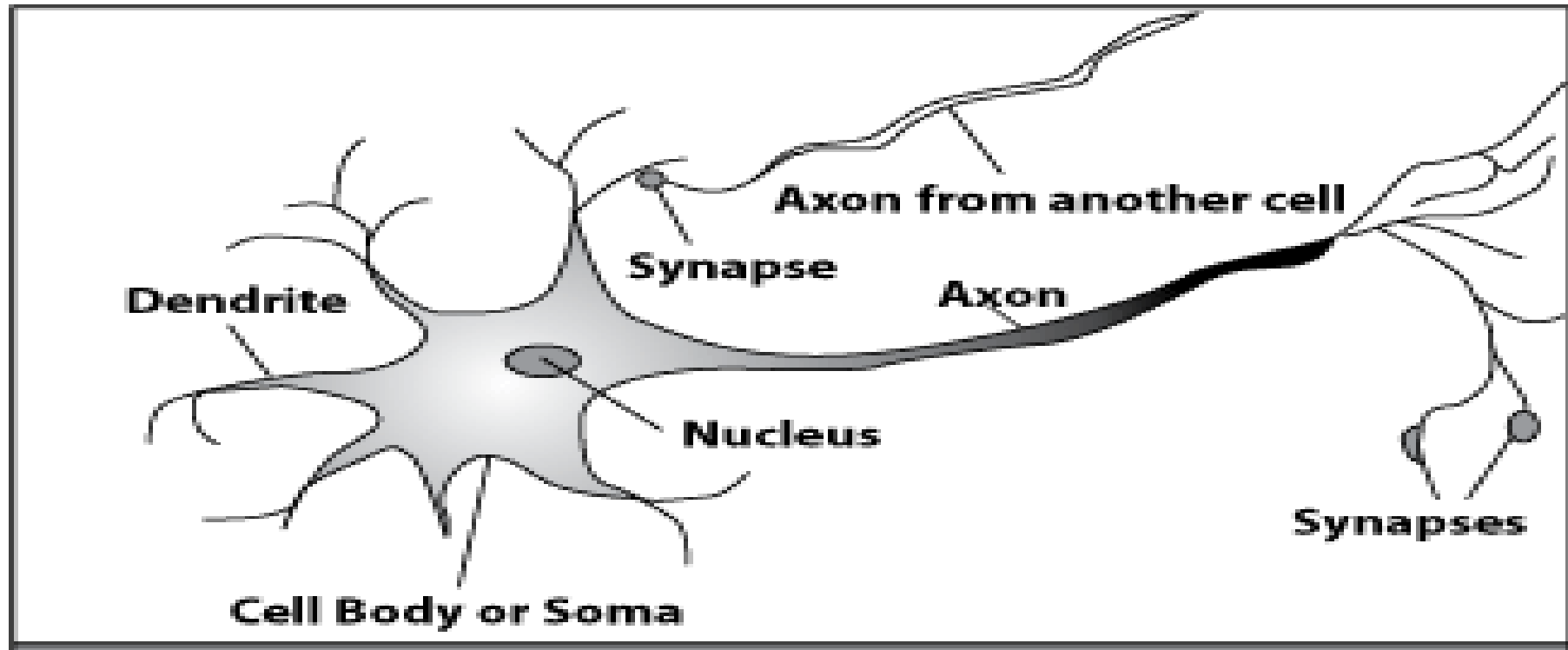
How do our brains work?

- The Brain is A massively parallel information processing system.
- Our brains are a huge network of processing elements. A typical brain contains a network of 10 billion neurons.



How do our brains work?

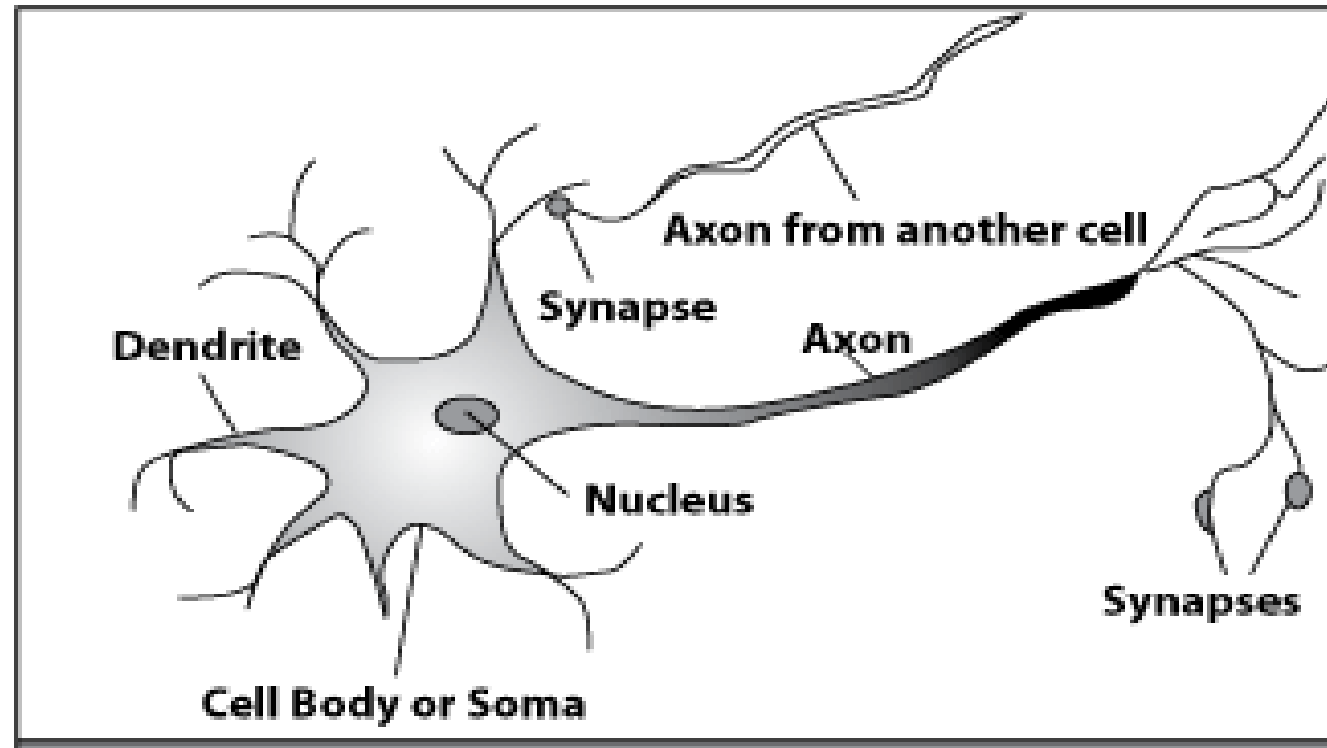
- A processing element



Dendrites: Input
Cell body: Processor
Synaptic: Link
Axon: Output

How do our brains work?

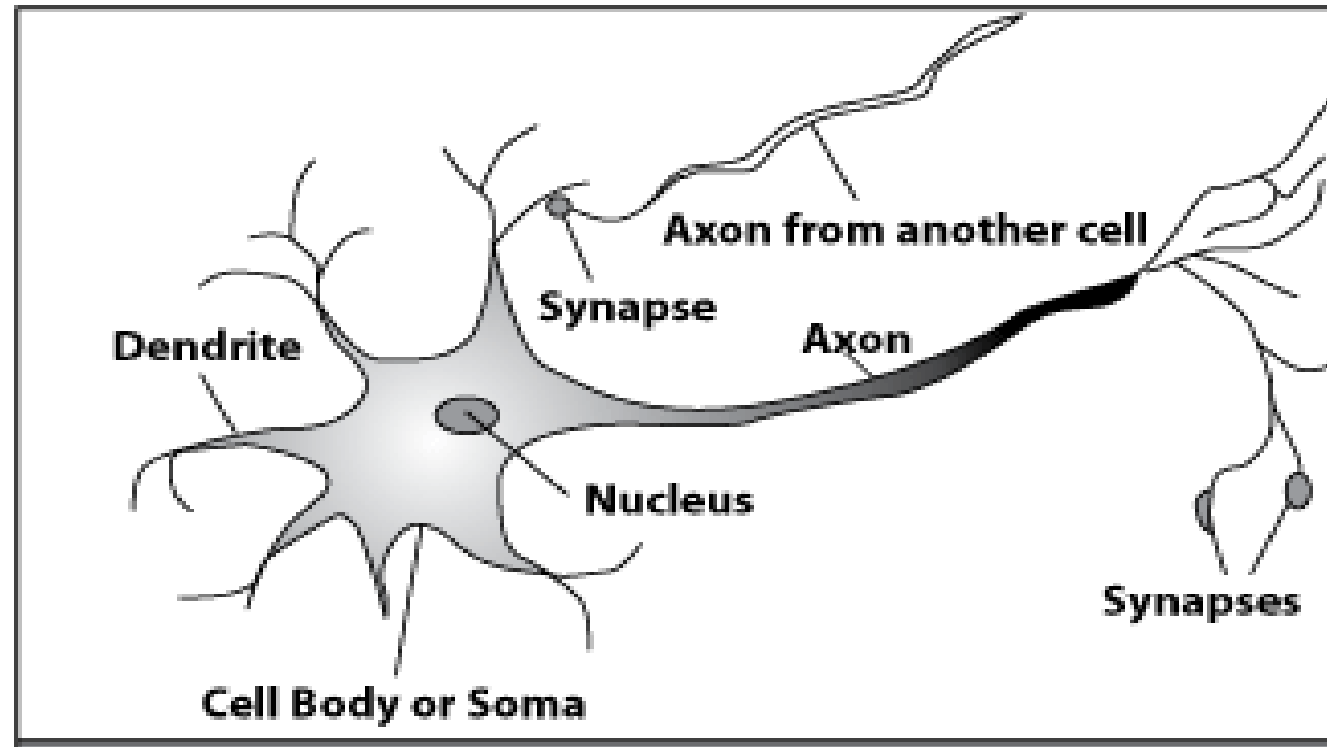
- A processing element



A neuron is connected to other neurons through about *10,000 synapses*

How do our brains work?

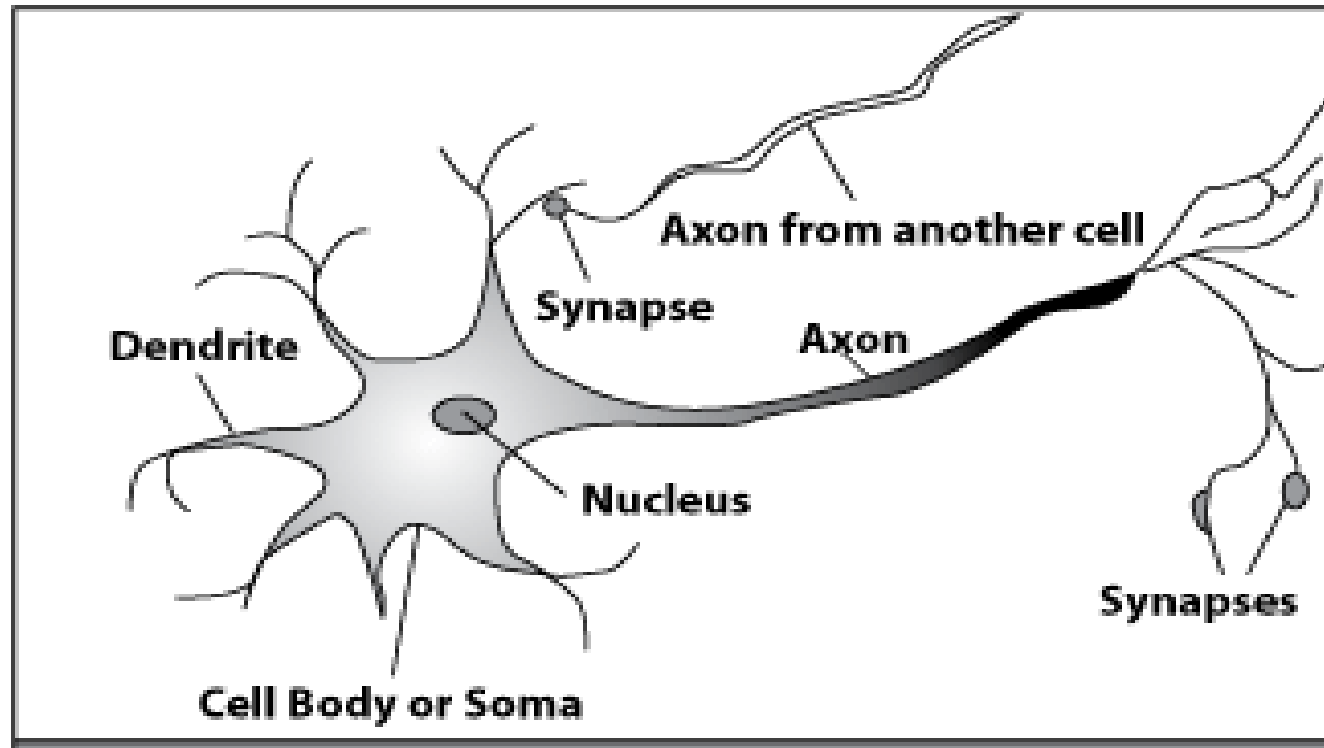
- A processing element



A neuron receives input from other neurons. Inputs are combined.

How do our brains work?

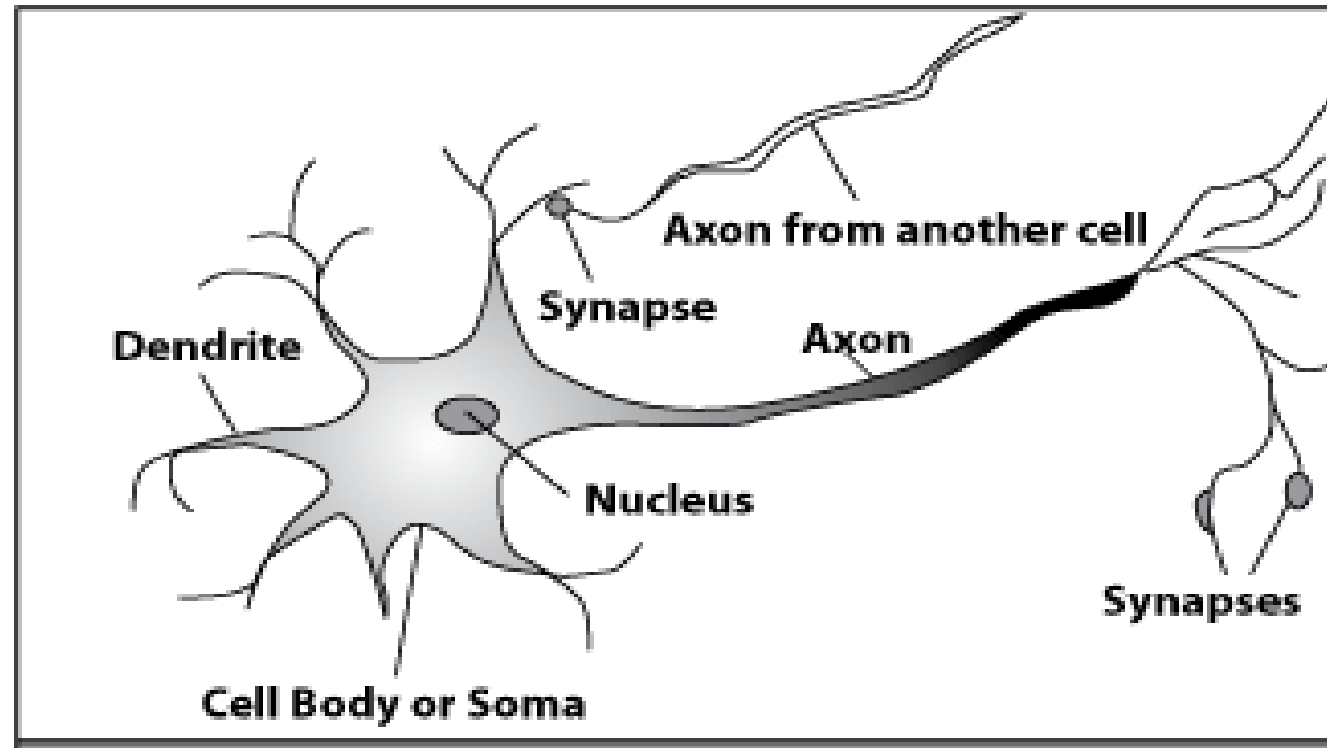
- A processing element



Once input exceeds a critical level, the neuron discharges a spike - an electrical pulse that travels from the body, down the axon, to the next neuron(s)

How do our brains work?

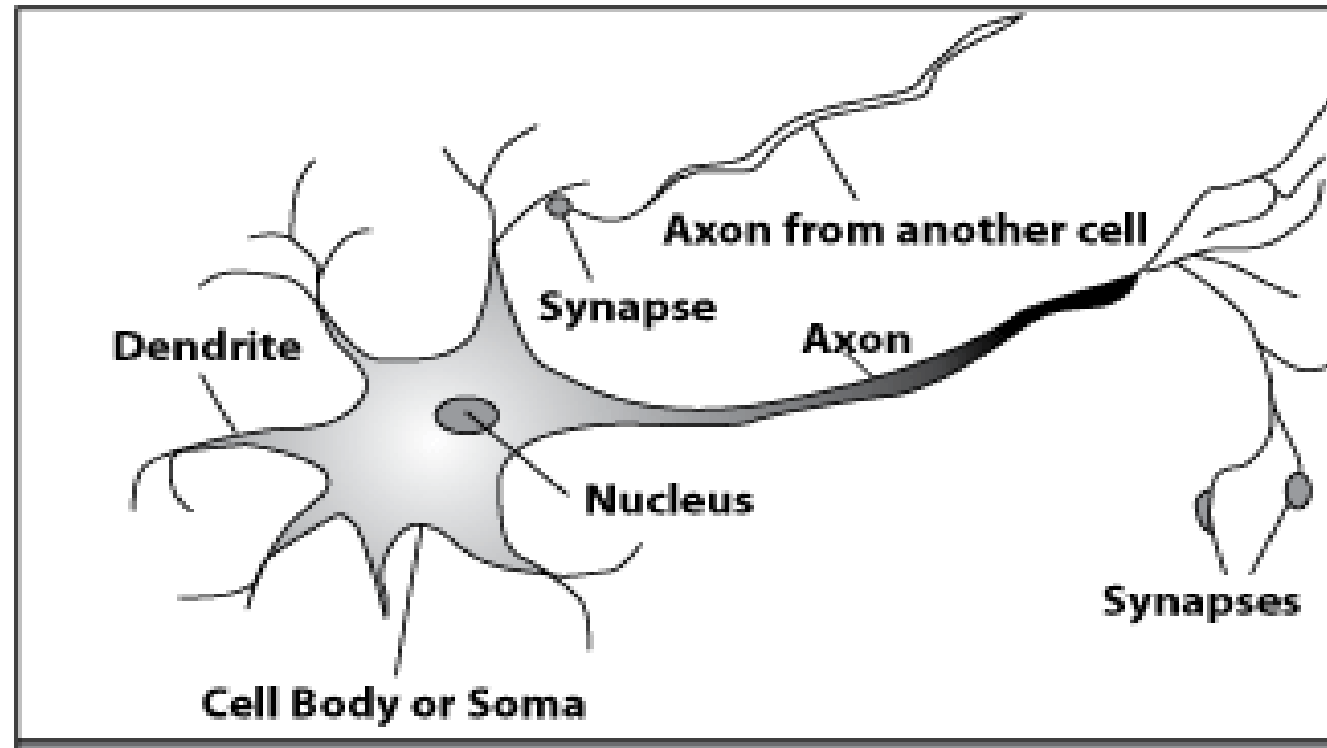
- A processing element



The axon endings almost touch the dendrites or cell body of the next neuron.

How do our brains work?

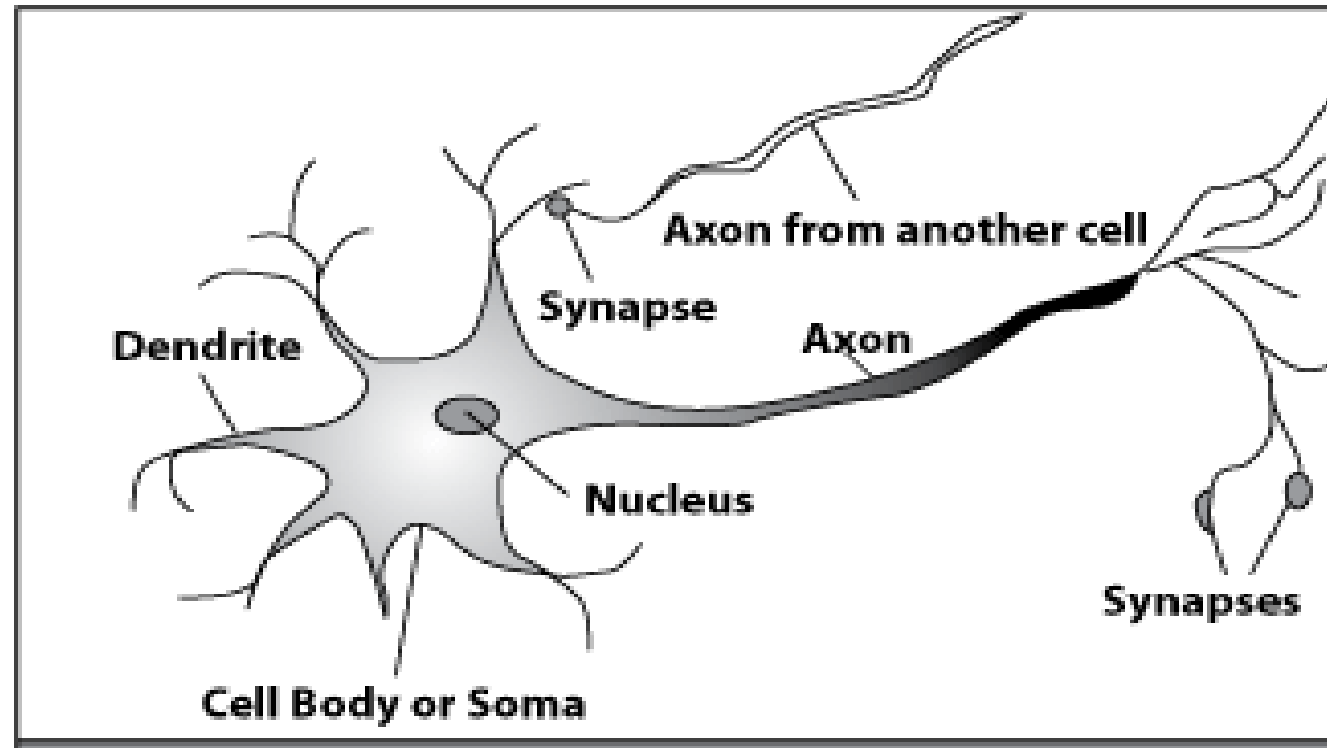
- A processing element



Transmission of an electrical signal from one neuron to the next is effected by neurotransmitters.

How do our brains work?

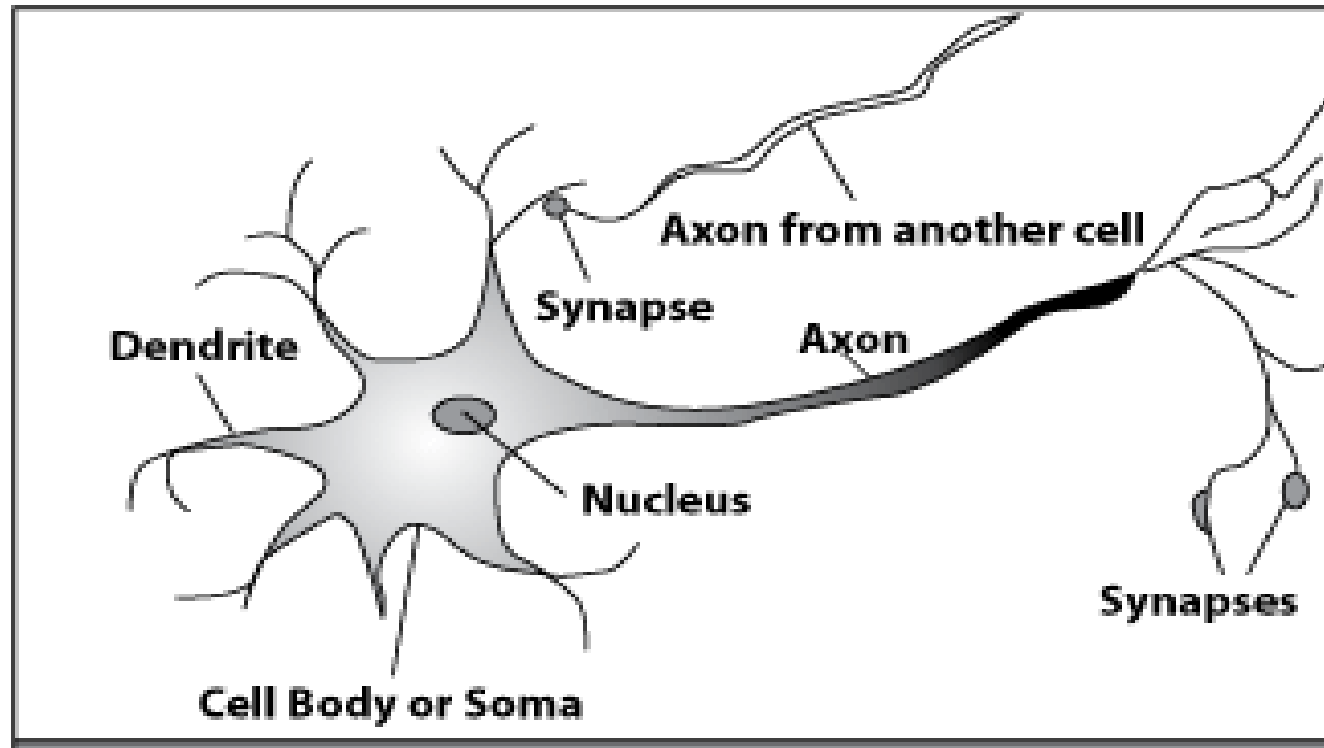
- A processing element



Neurotransmitters are chemicals which are released from the first neuron and which bind to the Second.

How do our brains work?

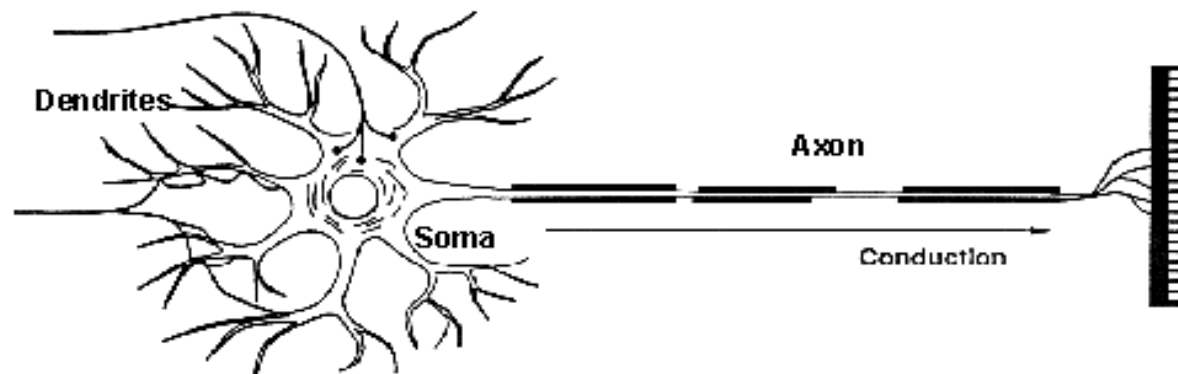
- A processing element



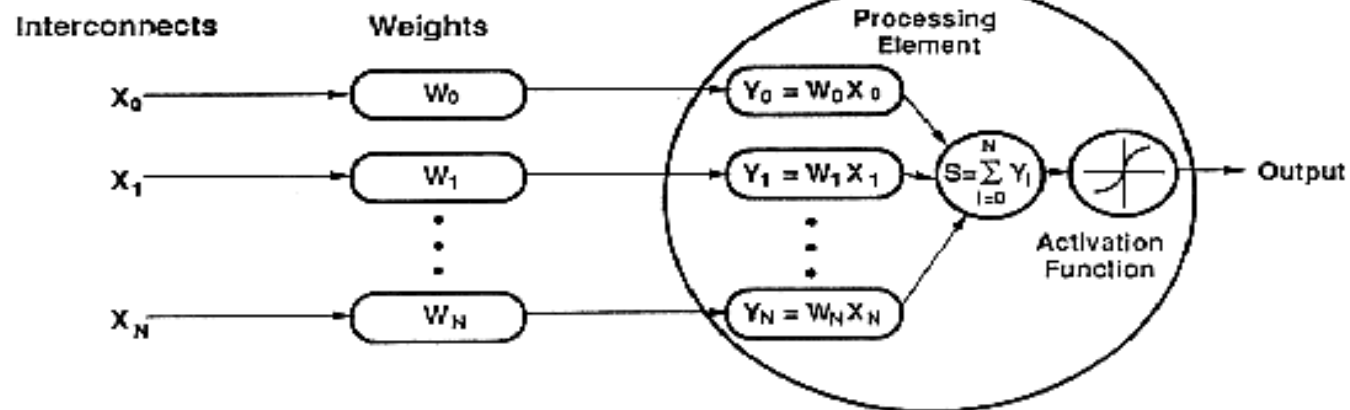
This link is called a synapse. The strength of the signal that reaches the next neuron depends on factors such as the amount of neurotransmitter available.

How do ANNs work?

Biological Neuron



Artificial Neuron

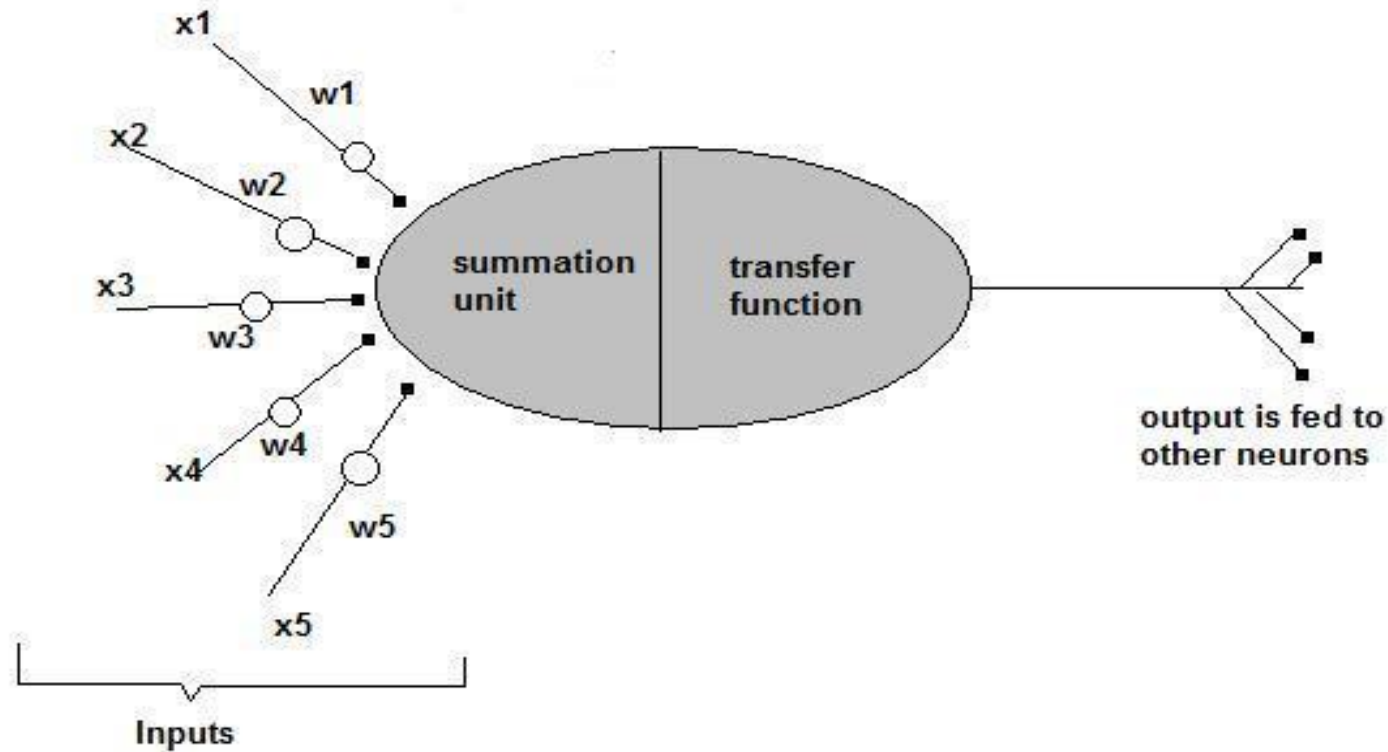


An artificial neuron is an imitation of a human neuron

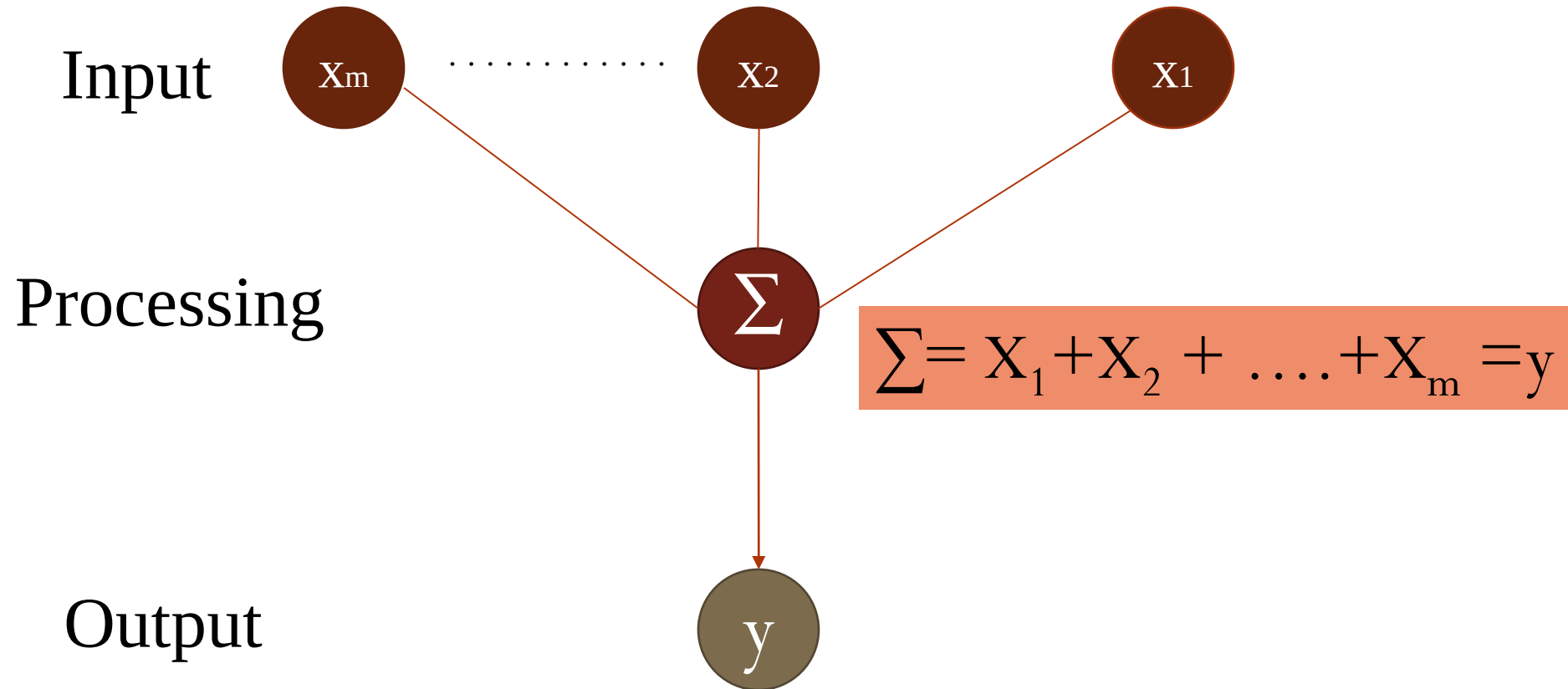
How do ANNs work?

- Now, let us have a look at the model of an artificial neuron.

A Single Neuron

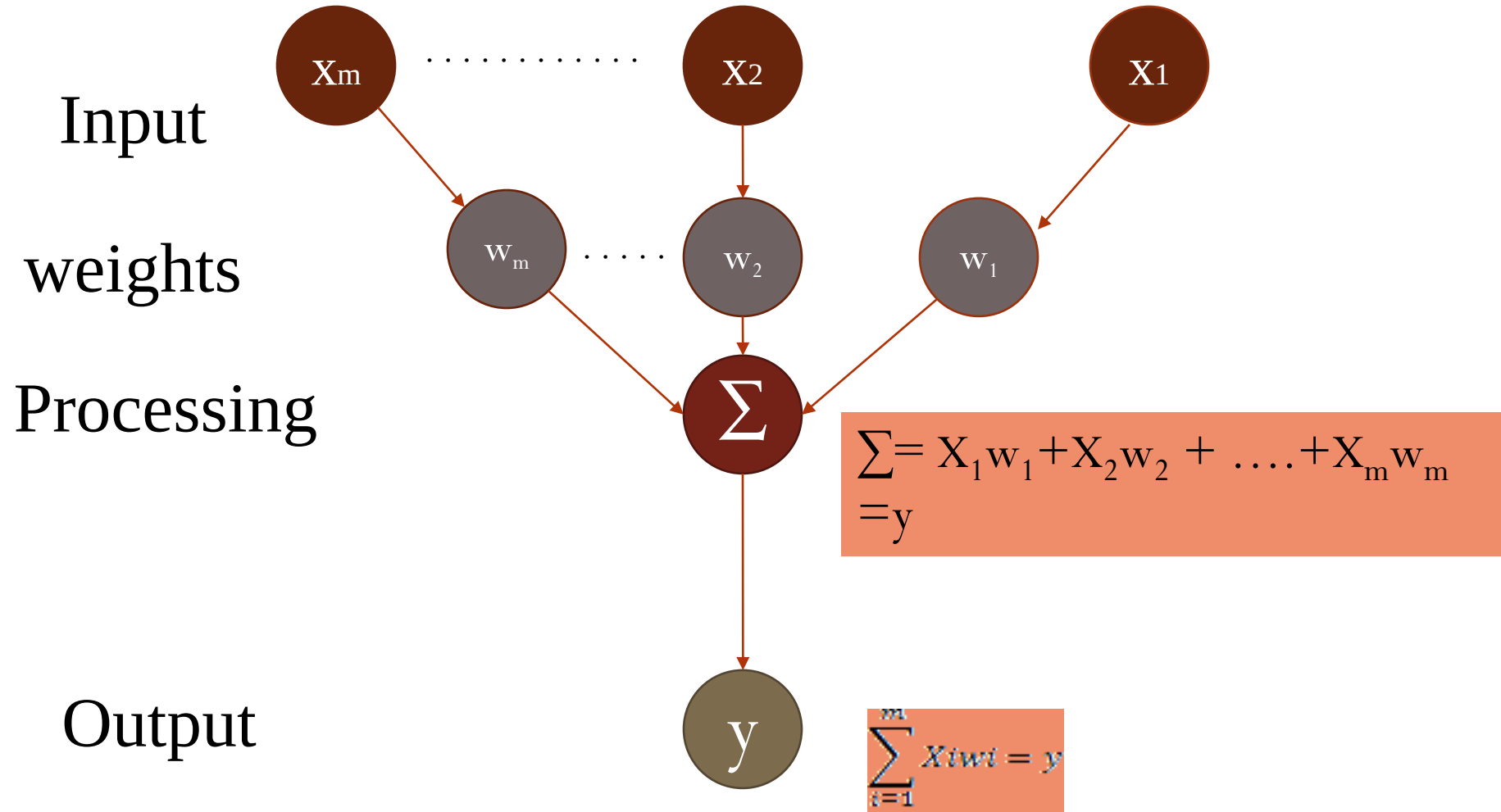


How do ANNs work?



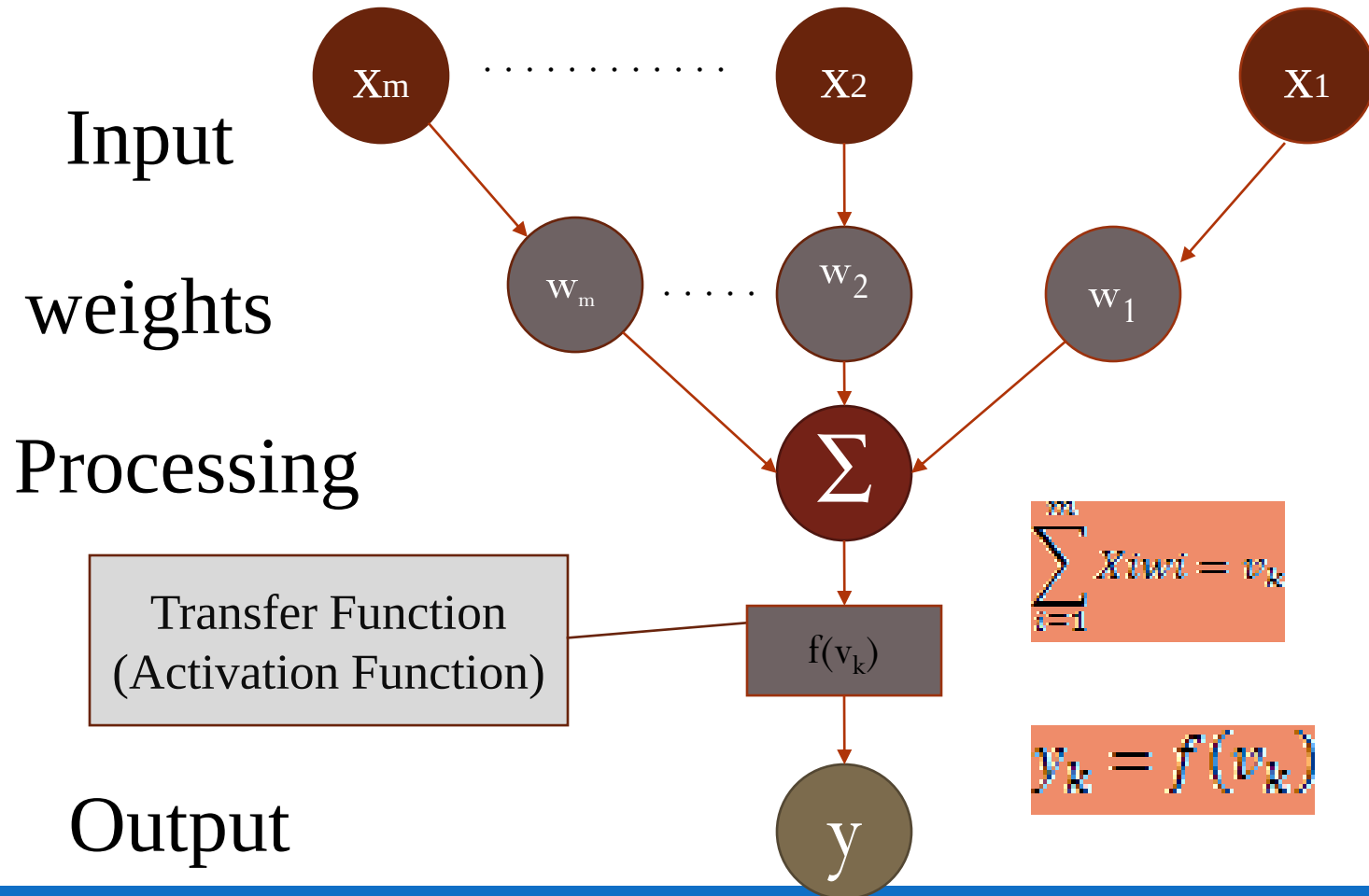
How do ANNs work?

Not all inputs are equal

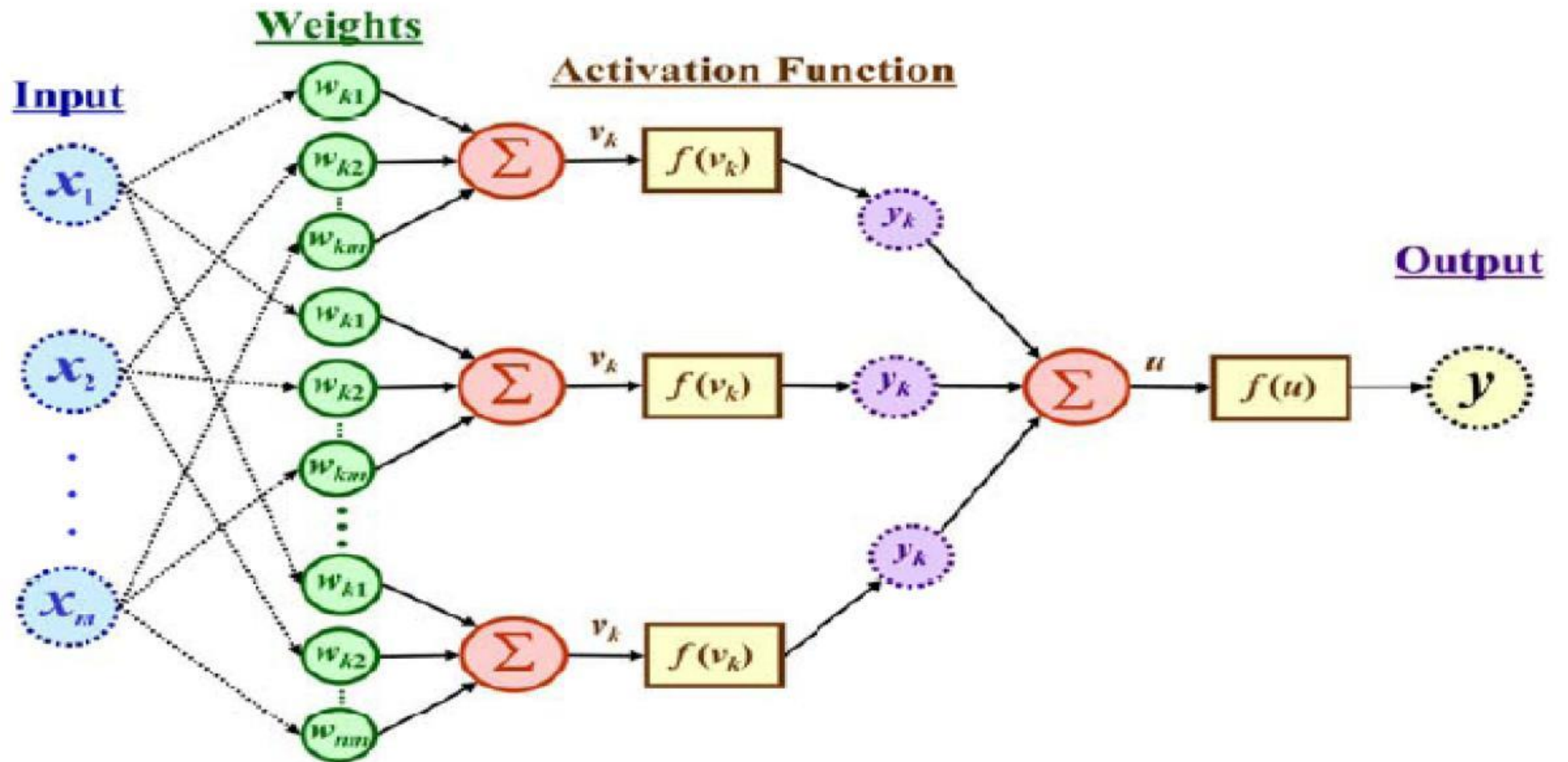


How do ANNs work?

The signal is not passed down to the next neuron verbatim



The output is a function of the input, that is affected by the weights, and the transfer functions



Artificial Neural Networks

- An ANN can:
 1. compute *any computable* function, by the appropriate selection of the network topology and weights values.
 2. learn from experience!
 - Specifically, by trial-and-error

Learning by trial-and-error

Continuous process of:

➤ Trial:

Processing an input to produce an output (In terms of ANN: Compute the output function of a given input)

➤ Evaluate:

Evaluating this output by comparing the actual output with the expected output.

➤ Adjust:

Adjust the *weights*.

NEURAL NETWORKS

-NEURAL NETWORK WAS INSPIRED BY THE DESIGN AND FUNCTIONING OF HUMAN BRAIN AND COMPONENTS.

-DEFINITION:

-INFORMATION PROCESSING MODEL THAT IS INSPIRED BY THE WAY BIOLOGICAL NERVOUS SYSTEM (I.E) THE BRAIN, PROCESS INFORMATION.

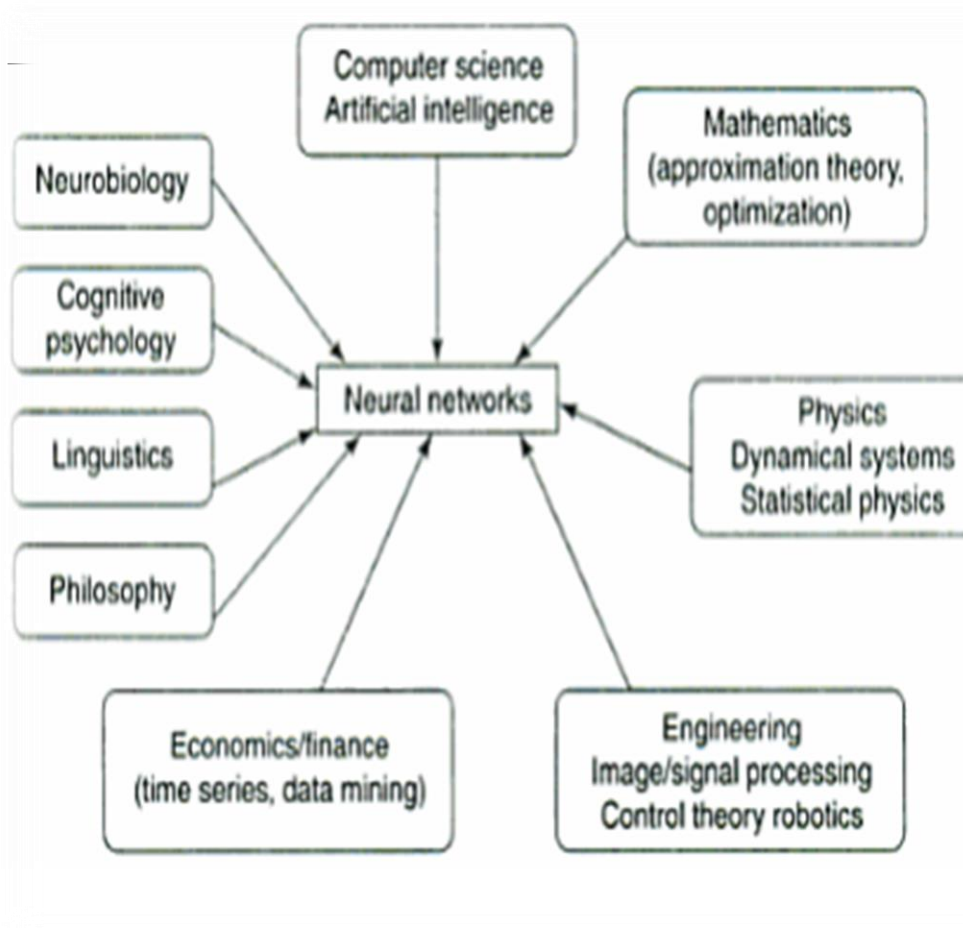
-ANN IS COMPOSED OF LARGE NUMBER OF HIGHLY INTERCONNECTED PROCESSING ELEMENTS(NEURONS) WORKING IN UNISON TO SOLVE PROBLEMS.

-IT IS CONFIGURED FOR SPECIAL APPLICATION SUCH AS PATTERN RECOGNITION AND DATA CLASSIFICATION THROUGH A LEARNING PROCESS.

-85-90% ACCURATE.

ADVANTAGES OF NEURAL NETWORKS

- A Neural Network can be an “*expert*” in analyzing the category of information given to it.
- Answers “ what-if” questions
- Adaptive learning
 - Ability to learn how to do tasks based on the data given for training or initial experience.
- Self organization
 - Creates its own organization or representation of information it receives during learning time.
- Real time operation
 - Computations can be carried out in parallel.
- Fault tolerance via redundant information coding
 - Partial destruction of neural network cause degradation of performance.
 - In some cases, it can be retained even after major network damage.
- In future, it can also used to give spoken words as instructions for machine.



This figure shows the multi disciplinary point of view of Neural Networks

APPLICATION SCOPE OF NEURAL NETWORKS

- | | |
|---|--------------------------------------|
| 1. Animal behavior | 12. Air traffic control |
| 2. Appraisal and valuation of property, etc., | 13. Fraud detection |
| 3. Betting on horse races, stock markets | 14. Hand writing and typewriting |
| 4. Criminal sentencing | 15. Lake water levels |
| 5. Complex physical and chemical process | 16. Machinery controls |
| 6. Data mining, cleaning and validation | 17. Medical diagnosis |
| 7. Direct mail advertisers | 18. Music composition |
| 8. Echo patterns | 19. Photos and finger prints |
| 9. Economic modeling | 20. Recipes and chemical formulation |
| 10. Employee hiring | 21. Traffic flows |
| 11. Expert consultants | 22. Weather prediction |

FUZZY LOGIC

Lofti Zadeh, Professor at University of California.

An organized method for dealing with imprecise data

Fuzzy logic includes 0 and 1 as extreme cases of truth (or "the state of matters" or "fact") but also includes the various states of truth in between so that, for example, the result of a comparison between two things could be not "tall" or "short" but ".38 of tallness."

Allows partial membership

Implemented in small, embedded micro controllers to large , networked, multichannel PC or work station.

Can be implemented in hardware, software or in both.

It mimics how a person would make decisions.

GENETIC ALGORITHM

How genes of parents combine to form those of their children.

Create an initial population of individuals representing possible solutions to solve a problem

Individual characters determine whether they are less or more fit to the population

The more fit members will take high probability.

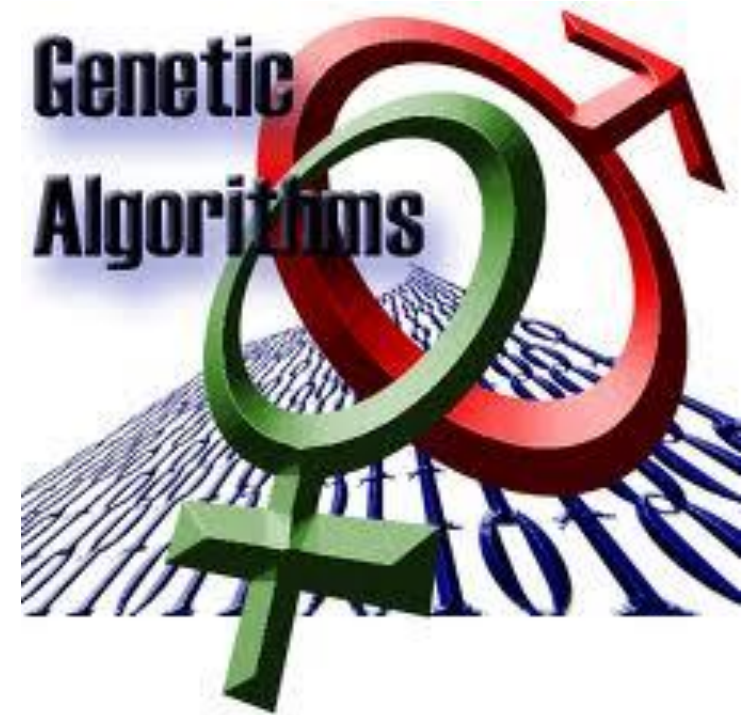
It is very effective in *finding optimal or near optimal solutions*.

Generate and test strategy.

Differ from normal optimization and search procedures in:

- Work with coding of the parameter set
- Work with multiple points
- Search via sampling(a blind search)
- Search using stochastic operators

In business, scientific and engineering circles, etc.,



HYBRID SYSTEM

Three types:-

Neuro Fuzzy hybrid system

- Combination of fuzzy set theory and neural networks
- Fuzzy system deal with explicit knowledge that can be explained and understood
- Neural network deal with implicit knowledge acquired by learning
- Advantages are:
 - Handle any kind of information
 - Manage imprecise, partial, vague or imperfect information
 - Resolve conflicts by collaboration and aggregation.
 - Self learning, self organizing and self tuning capability
 - No need for prior knowledge of relationship of data
 - Mimic human decision making system

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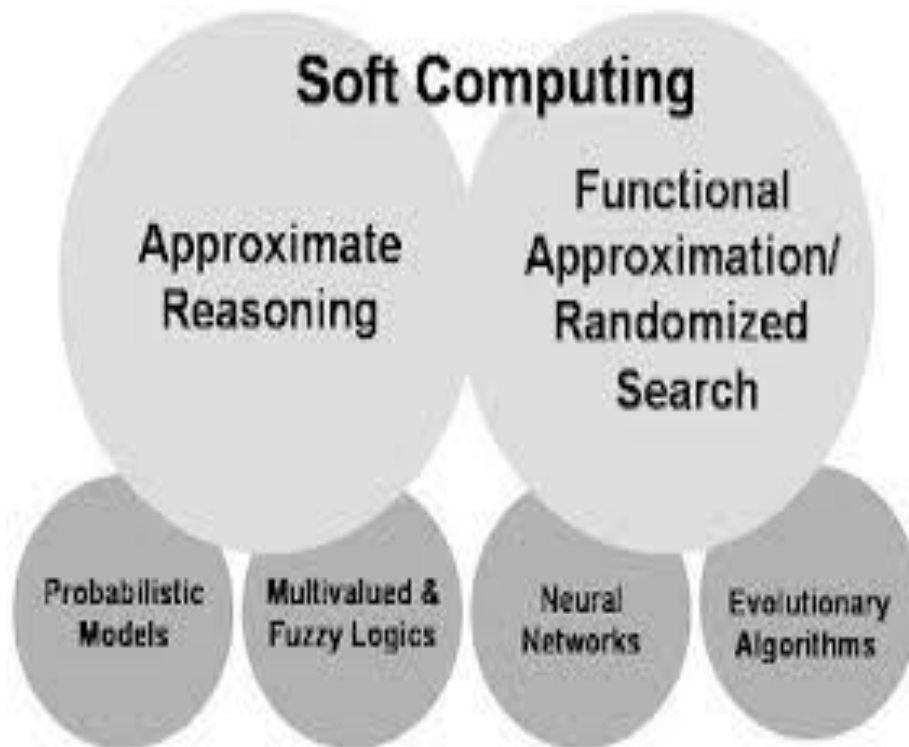
Neuron genetic hybrid system

- Topology optimization
 - Select a topology for ANN , common one is back propagation
- Genetic algorithm training
 - Learning of ANN is formulated as weight optimization problem, usually mean squared error as a fitness measure
- Control parameter optimization
 - Learning rate, momentum rate, tolerance level. Etc., are optimized using GA.

Fuzzy genetic hybrid system

- Creating the classification rules for a fuzzy system where objects are classified by linguistic terms.

SOFT COMPUTING



Two major problem solving techniques are:

- Hard computing
 - Deals with precise model where accurate solutions are achieved.
- Soft computing
 - deals with approximate model to give solution for complex problems
 - Prof. Lotfi Zadeh introduced it.
 - Ultimate goal-emulate the human mind
 - It is a ***combination of GA, Neural Network and FL.***

ARTIFICIAL NEURAL NETWORK : AN INTRODUCTION

- Resembles the characteristic of biological neural network.
- **Nodes** – interconnected processing elements (units or neurons)
- Neuron is connected to other by a **connection link**.
- Each connection link is associated with **weight** which has information about the input signal.
- ANN processing elements are called as **neurons or artificial neurons** , since they have the capability to model networks of original neurons as found in brain.
- Internal state of neuron is called **activation or activity level** of neuron, which is the function of the inputs the neurons receives.
- Neuron can send only one signal at a time.

BASIC OPERATION OF A NEURAL NET

X1 and X2 – input neurons.

Y- output neuron

Weighted interconnection links- W1 and W2.

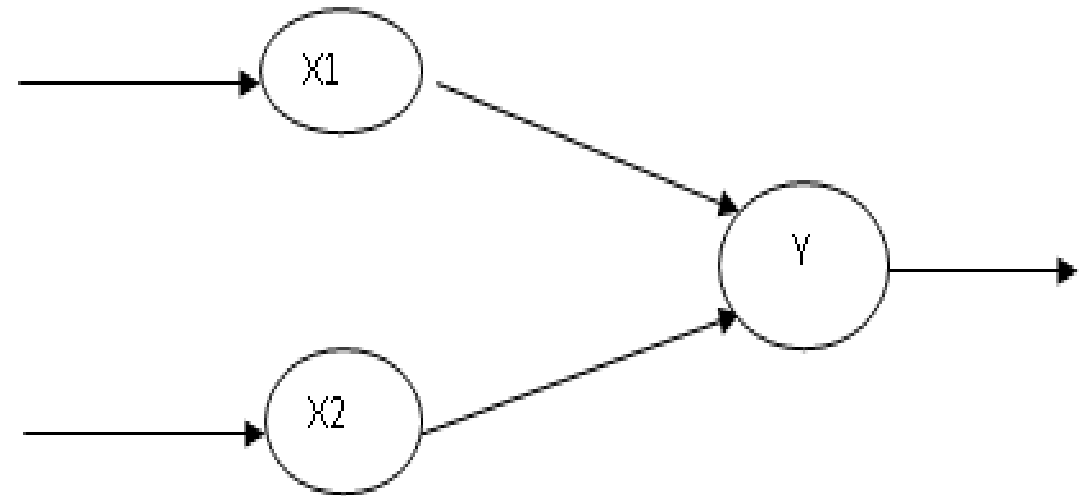
Net input calculation is :

Output is :

$$Y_{in} = -x_1w_1 + x_2w_2$$

Output= function

$$y = f(Y_{in})$$



CONTD...

The function to be applied over the net input is called ***activation function***.

Weight involved in ANN is equal to the slope of linear straight line ($y=mx$).

BIOLOGICAL NEURAL NETWORK

Has three main parts

- Soma or cell body-where cell nucleus is located
 - Dendrites-where the nerve is connected to the cell body
 - Axon-which carries the impulses of the neuron
-

Electric impulse is passed between synapse and dendrites.

Synapse- Axon split into strands and strands terminates into small bulb like organs called as synapse.

It is a chemical process which results in increase /decrease in the electric potential inside the body of the receiving cell.

If the electric potential reaches a thresh hold value, receiving cell fires & pulse / action potential of fixed strength and duration is send through the axon to synaptic junction of the cell.

After that, cell has to wait for a period called *refractory period*.

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In this model net input is calculated by

$$Y_{in} = x_1w_1 + x_2w_2 + \dots + x_nw_n$$

$$\sum_{i=1}^n x_iw_i$$

Terminology Relation Between Biological And Artificial Neuron

Biological Neuron	Artificial Neuron
Cell	Neuron
Dendrites	Weights or interconnections
Soma	Net input
Axon	Output

CHARACTERISTICS OF ANN:

Neurally implemented mathematical model

Large number of processing elements called neurons exists here.

Interconnections with weighted linkage hold informative knowledge.

Input signals arrive at processing elements through connections and connecting weights.

Processing elements can learn, recall and generalize from the given data.

Computational power is determined by the collective behavior of neurons.

- *ANN is a connection models, parallel distributed processing models, self-organizing systems, neuro-computing systems and neuro morphic system.*

BASIC MODELS OF ANN

Models are based on three entities

- The model's synaptic interconnections.
- The training or learning rules adopted for updating and adjusting the connection weights.
- Their activation functions

The *arrangement of neurons to form layers* and the *connection pattern formed within and between layers* is called the **network architecture**.

Five types:

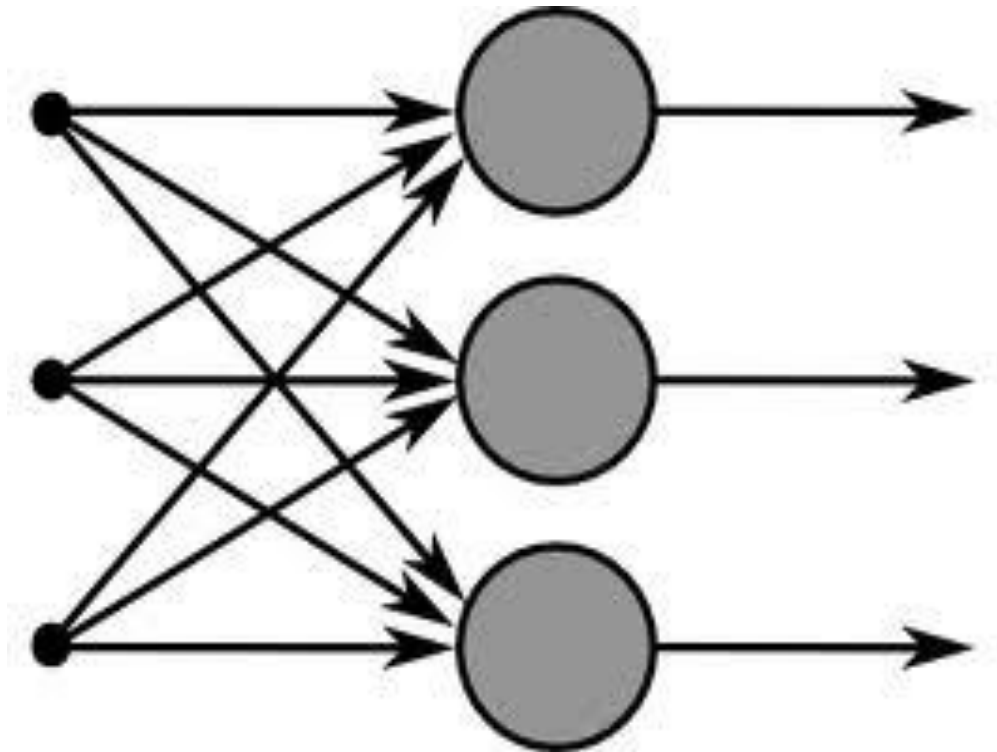
- Single layer feed forward network
- Multilayer feed-forward network
- Single node with its own feedback
- Single-layer recurrent network
- Multilayer recurrent network

Single layer Feed- Forward Network

Layer is formed by taking processing elements and combining it with other processing elements.

Input and output are linked with each other

Inputs are connected to the processing nodes with various weights, resulting in series of outputs one per node.



Multilayer feed-forward network

Formed by the interconnection of several layers.

Input layer receives input and buffers input signal.

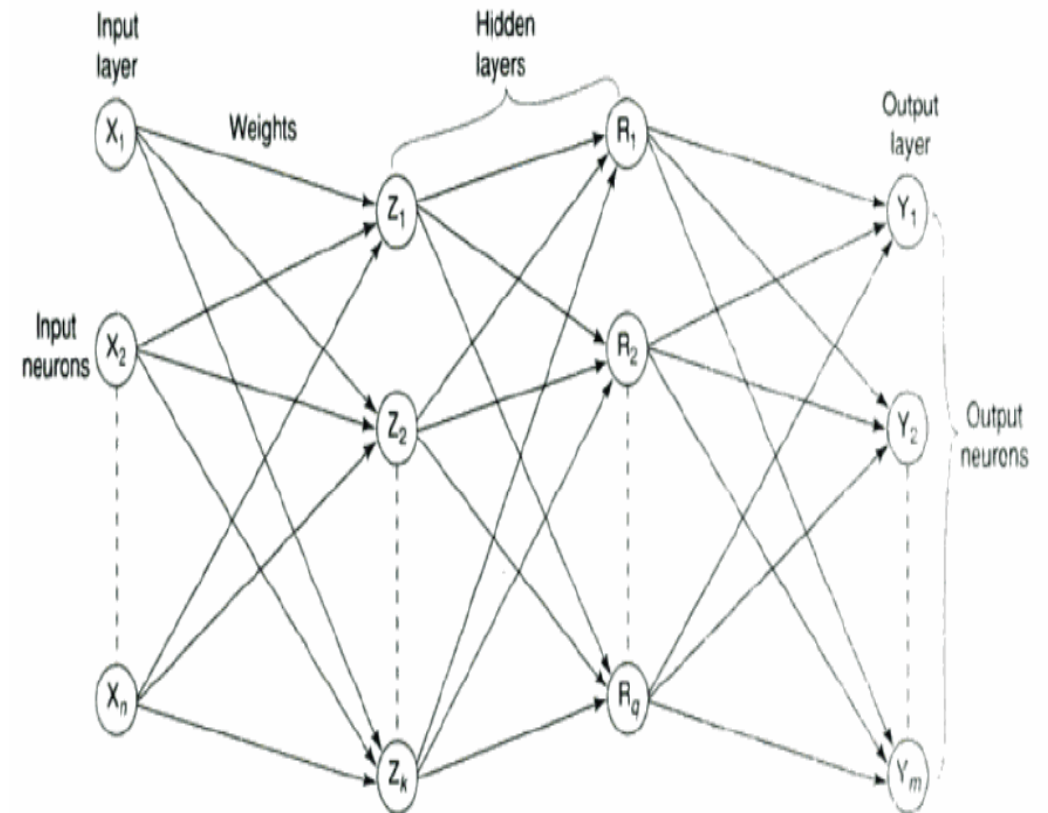
Output layer generated output.

Layer between input and output is called *hidden layer*.

Hidden layer is internal to the network.

Zero to several hidden layers in a network.

More the hidden layer, more is the complexity of network, but efficient output is produced.



FEED BACK NETWORK

If no neuron in the output layer is an input to a node in the same layer / proceeding layer – **feed forward network**.

If outputs are directed back as input to the processing elements in the same layer/proceeding layer – **feedback network**.

If the output are directed back to the input of the same layer then it is **lateral feedback**.

Recurrent networks are networks with feedback networks with closed loop.

Fig 2.8 (A) –simple recurrent neural network having a single neuron with feedback to itself.

Fig 2.9 – single layer network with feedback from output can be directed to processing element itself or to other processing element/both.

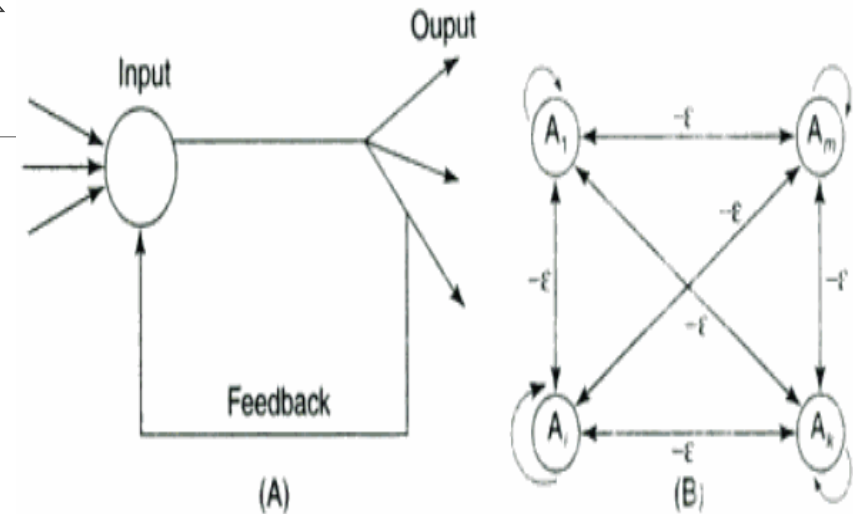


Figure 2-8 (A) Single node with own feedback. (B) Competitive nets.

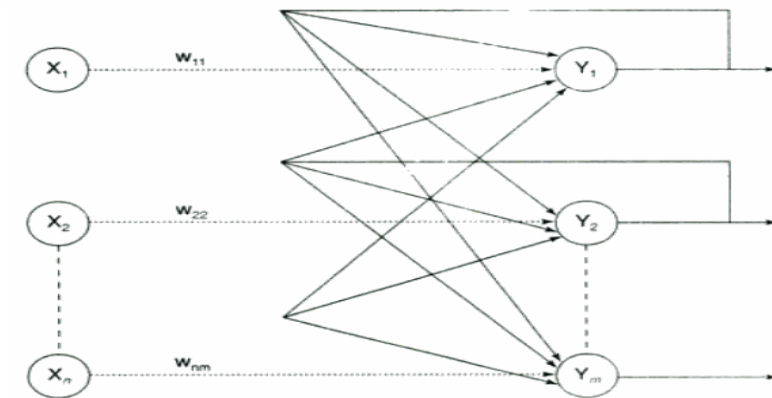
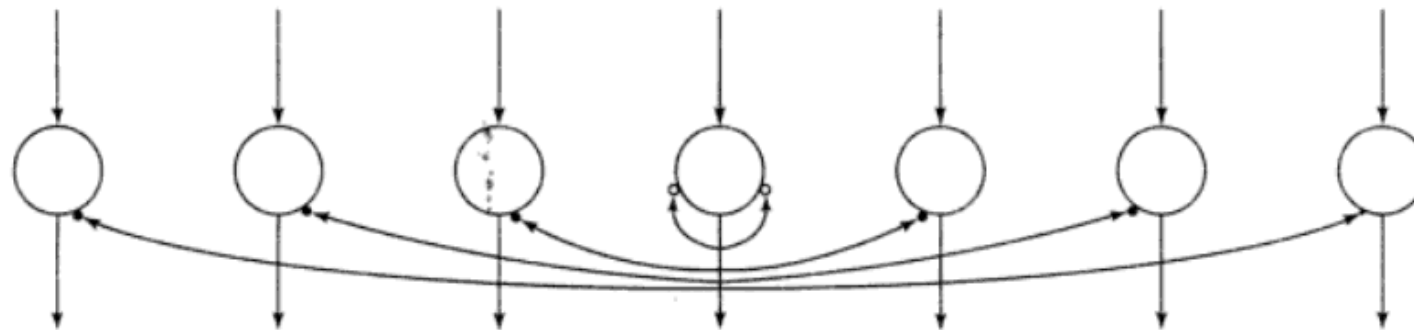
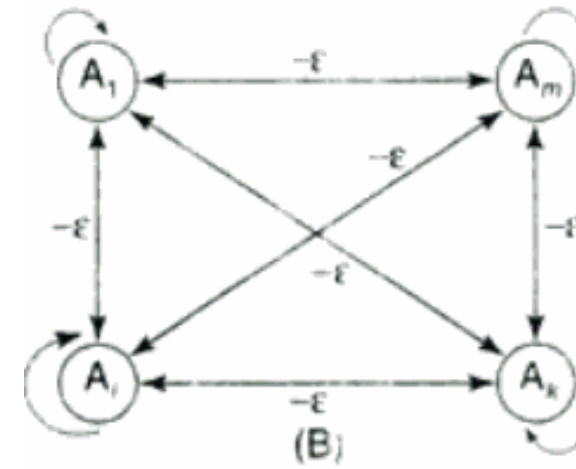


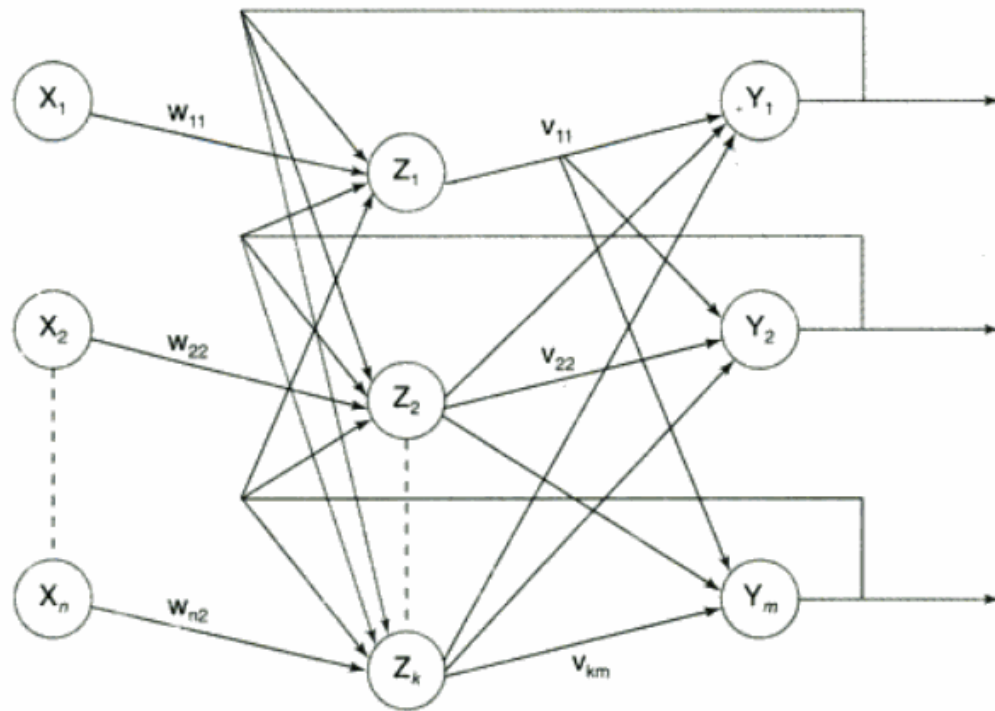
Figure 2-9 Single-layer recurrent network.

Maxnet –competitive interconnections having fixed weights.

On-center-off-surround/lateral inhibition structure – each processing neuron receives two different classes of inputs- “*excitatory*” input from nearby processing elements & “*inhibitory*” elements from more distantly located precessing elements. This type of interconnection is shown below



Multilayer Recurrent Network



Processing element output can be directed back to the nodes in the preceding layer, forming a ***multilayer recurrent network***.

Processing element output can be directed to processing element itself or to other processing element in the same layer.

LEARNING

Two broad kinds of learning in ANNs is :

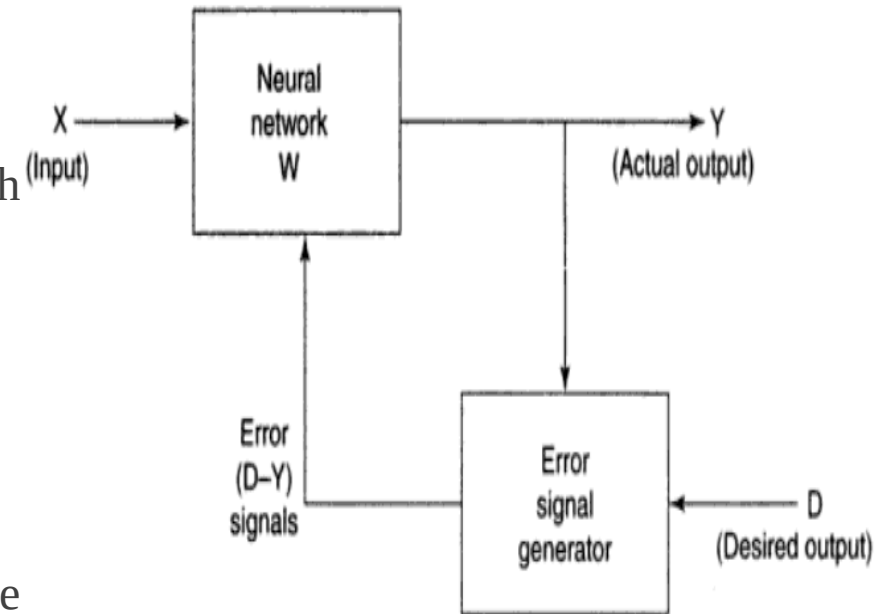
- i) parameter learning – updates connecting weights in a neural net.
- ii) Structure learning – focus on change in the network.

Apart from these, learning in ANN is classified into three categories as

- i) supervised learning
- ii) unsupervised learning
- iii) reinforcement learning

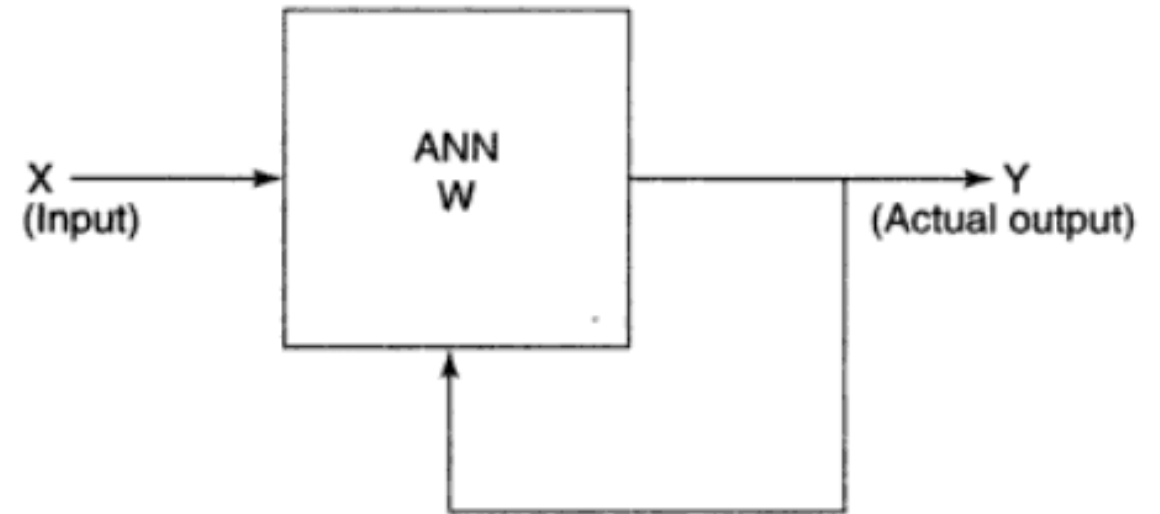
SUPERVISED LEARNING

- Learning with the help of a teacher.
- Example : learning process of a small child.
 - Child doesn't know read/write.
 - Their each & every action is supervised by a teacher
- In ANN, each input vector requires a corresponding target vector, which represents the desired output.
- The input vector along with target vector is called **training pair**.
- The input vector results in output vector.
- The actual output vector is compared with desired output vector.
- If there is a difference means an error signal is generated by the network.
- It is used for adjustment of weights until actual output matches desired output.

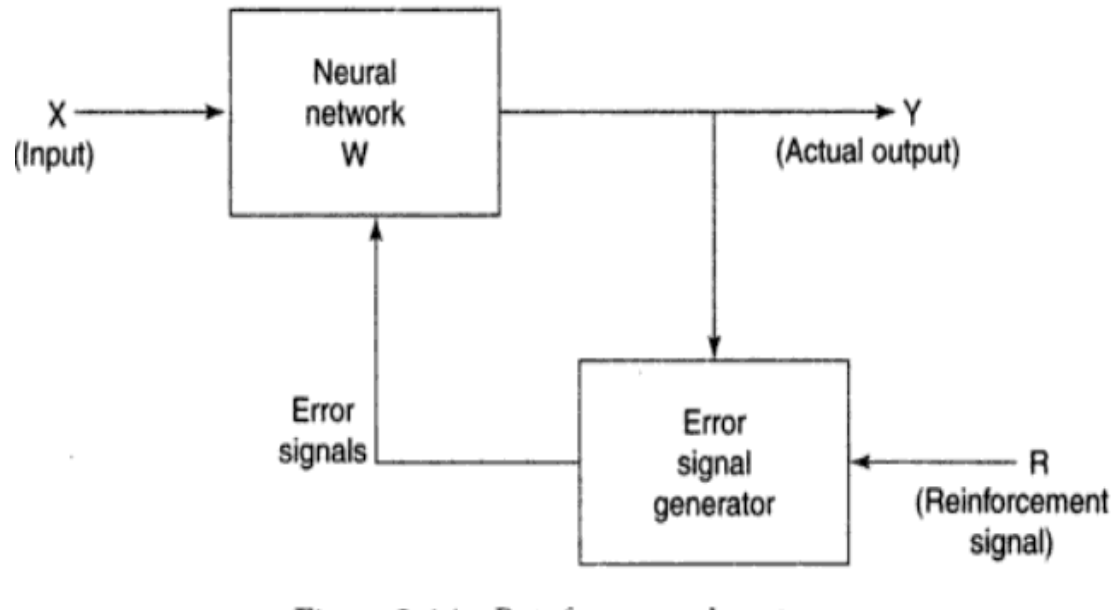


UNSUPERVISED LEARNING

- Learning is performed without the help of a teacher.
- Example: tadpole – learn to swim by itself.
- In ANN, during training process, network receives input patterns and organize it to form clusters.
- From the Fig. it is observed that no feedback is applied from environment to inform what output should be or whether they are correct.
- The network itself discover patterns, regularities, features/ categories from the input data and relations for the input data over the output.
- Exact clusters are formed by discovering similarities & dissimilarities so called as *self – organizing*.



REINFORCEMENT LEARNING



- Similar to supervised learning.
- Learning based on *critic information* is called *reinforcement learning* & the feedback sent is called *reinforcement signal*.
- The network receives some feedback from the environment.
- Feedback is only evaluative.
- The external reinforcement signals are processed in the critic signal generator, and the obtained critic signals are sent to the ANN for adjustment of weights properly to get critic feedback in future.

ACTIVATION FUNCTIONS

- To make work more efficient and for exact output, some force or activation is given.
- Like that, activation function is applied over the net input to calculate the output of an ANN.
- Information processing of processing element has two major parts: input and output.
- An integration function (f) is associated with input of processing element.
- Several activation functions are there.

1. Identity function:

- it is a linear function which is defined as

$$f(x) = x \text{ for all } x$$

- The output is same as the input.

2. Binary step function

- it is defined as

$$f(x) = \begin{cases} 1 & \text{if } x \geq \theta \\ 0 & \text{if } x < \theta \end{cases}$$

where θ represents thresh hold value.

It is used in single layer nets to convert the net input to an output that is bianary. (0 or 1)

CONTD..

3. Bipolar step function:

- It is defined as

$$f(x) = \begin{cases} 1 & \text{if } x \geq \theta \\ -1 & \text{if } x < \theta \end{cases}$$

- where θ represents threshold value.
- used in single layer nets to convert the net input to an output that is bipolar (+1 or -1).

4. Sigmoid function

used in Back propagation nets.

Two types:

a) binary sigmoid function

-logistic sigmoid function or unipolar sigmoid function.

-it is defined as
$$f(x) = \frac{1}{1 + e^{-\lambda x}}$$

where λ – steepness parameter.

-The derivative of this function is

$f'(x) = \lambda f(x)[1-f(x)]$. The range of sigmoid function is 0 to 1.

CONTD..

b) *Bipolar sigmoid function*

$$f(x) = \frac{2}{1 + e^{-\lambda x}} - 1 = \frac{1 - e^{-\lambda x}}{1 + e^{-\lambda x}}$$

where λ - steepness parameter and the sigmoid range is between -1 and +1.

- The derivative of this function can be

$$[1 + f(x)][1 - f(x)]$$

$$f'(x) = \frac{\lambda}{2}$$

- It is closely related to hyperbolic tangent function, which is written as

$$h(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$h(x) = \frac{1 - e^{-2x}}{1 + e^{-x}}$$

CONTD..

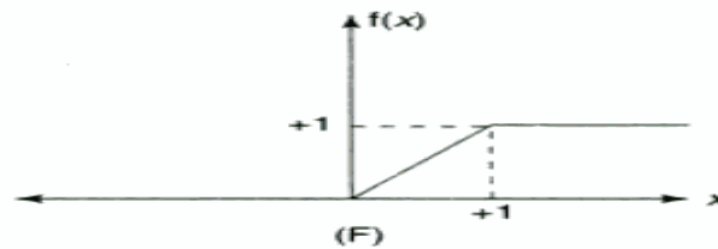
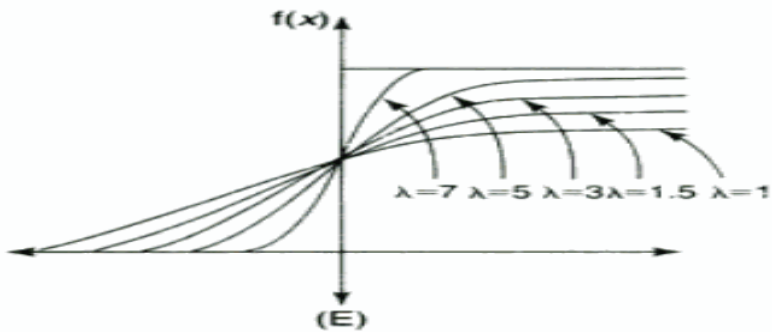
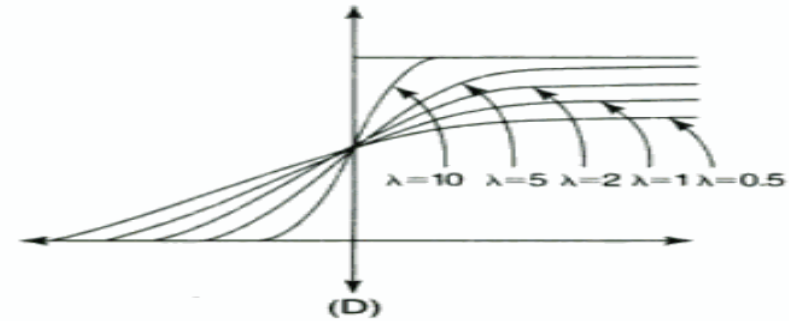
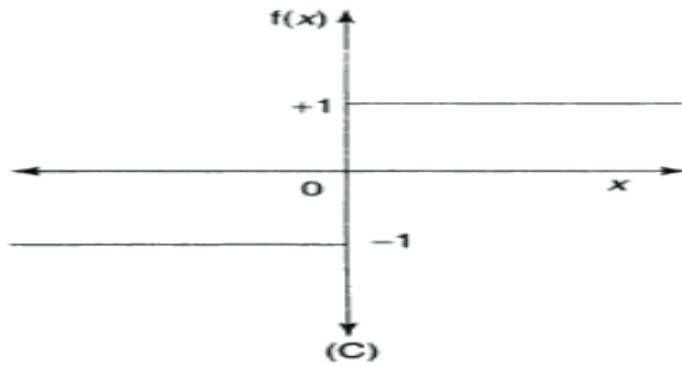
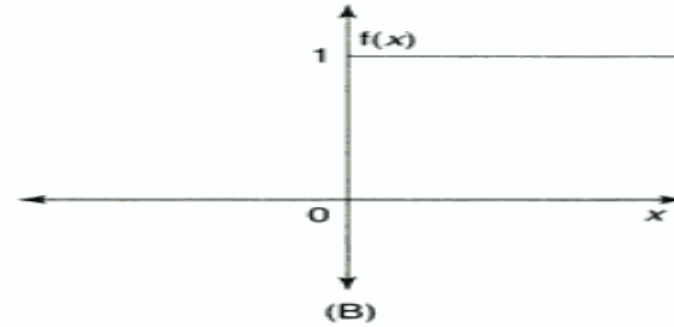
The derivative of the hyperbolic tangent function is

$$h'(x) = [1+h(x)][1-h(x)]$$

5. Ramp function

$$f(x) = \begin{cases} 1 & \text{if } x > 1 \\ x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{if } x < 0 \end{cases}$$

The graphical representation of all these functions is given in the upcoming Figure



2-15 Depiction of activation functions: (A) identity function; (B) binary step function; (C) bipolar step function; (D) binary sigmoidal function; (E) bipolar sigmoidal function; (F) ramp function.

IMPORTANT TERMINOLOGIES

- **Weight**

- The weight contain information about the input signal.
- It is used by the net to solve the problem.
- It is represented in terms of matrix & called as *connection matrix*.
- If weight matrix W contains all the elements of an ANN, then the set of all W matrices will determine the set of all possible information processing configuration.
- The ANN can be realized by finding an appropriate matrix W .
- Weight encode long-term memory (LTM) and the activation states of network encode short-term memory (STM) in a neural network.

CONTD..

Bias

- Bias has an impact in calculating net input.
- Bias is included by adding x_0 to the input vector x .
- The net output is calculated by

$$y_{in_j} = \sum_{i=0}^n x_i w_{ij}$$

$$y_{in_j} = b_j + \sum_{i=0}^n x_i w_{ij}$$

- The bias is of two types
 - Positive bias
 - Increase the net input
 - Negative bias
 - Decrease the net input

CONTD..

Threshold

- It is a set value based upon which the final output is calculated.
- Calculated net input and threshold is compared to get the network output.
- The activation function of threshold is defined as
 - where θ is the fixed threshold value

$$f(net) = \begin{cases} 1 & \text{if } net \geq \theta \\ -1 & \text{if } net < \theta \end{cases}$$

CONTD..

- **Learning rate**

- Denoted by α .
- Control the amount of weight adjustment at each step of training.
- The learning rate range from 0 to 1.
- Determine the rate of learning at each step

- **Momentum Factor**

- Convergence is made faster if a momentum factor is added to the weight updation process.
- Done in back propagation network.

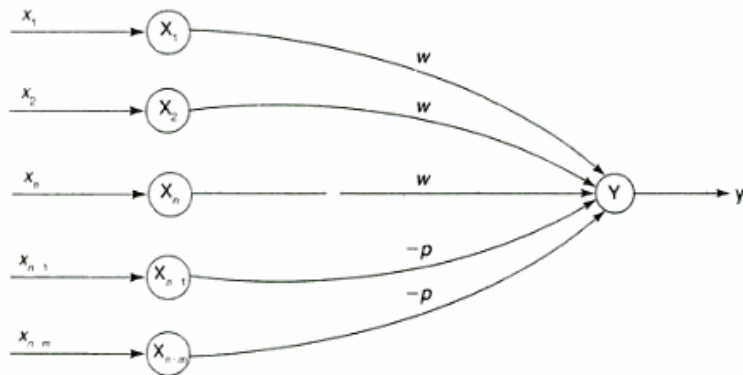
- **Vigilance parameter**

- Denoted by ρ .
- Used in Adaptive Resonance Theory (ART) network.
- Used to control the degree of similarity.
- Ranges from 0.7 to 1 to perform useful work in controlling the number of clusters.

MCCULLOCH-PITTS NEURON

- Discovered in 1943.
- Usually called as *M-P neuron*.
- M-P neurons are connected by directed weighted paths.
- Activation of M-P neurons is binary (i.e) at any time step the neuron may fire or may not fire.
- Weights associated with communication links may be excitatory(wgts are positive)/inhibitory(wgts are negative).
- Threshold plays major role here. There is a fixed threshold for each neuron and if the net input to the neuron is greater than the threshold then the neuron fires.
- They are widely used in logic functions.

CONTD...



- A simple M-P neuron is shown in the figure.
- It is excitatory with weight ($w > 0$) / inhibitory with weight $-p$ ($p < 0$).
- In the Fig., inputs from x_1 to x_n possess excitatory weighted connection and x_{n+1} to x_{n+m} has inhibitory weighted interconnections.
- Since the firing of neuron is based on threshold, activation function is defined as

$$f(y_{in}) = \begin{cases} 1 & \text{if } y_{in} \geq \theta \\ 0 & \text{if } y_{in} < \theta \end{cases}$$

CONTD...

- For inhibition to be absolute, the threshold with the activation function should satisfy the following condition:

$$\theta > \sum w_i - p$$

- Output will fire if it receives “ k ” or more excitatory inputs but no inhibitory inputs where

$$\sum_{i=1}^k w_i \geq \theta > \sum_{i=1}^{k-1} w_i$$

- The M-P neuron has no particular training algorithm.
- An analysis is performed to determine the weights and the threshold.
- It is used as a building block where any function or phenomenon is modeled based on a logic function.

LINEAR SEPARABILITY

- It is a concept wherein the separation of the input space into regions is based on whether the network response is positive or negative.
- A decision line is drawn to separate positive or negative response.
- The decision line is also called as *decision-making line* or *decision-support line* or *linear-separable line*.
- The net input calculation to the output unit is given as

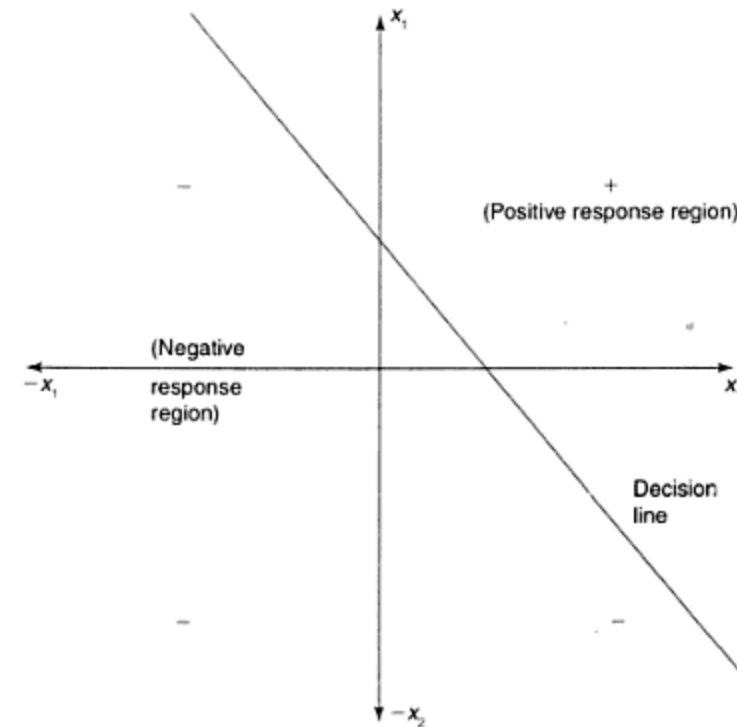
$$y_{in} = b_j + \sum_{i=1}^n x_i w_i$$

- The region which is called as *decision boundary* is determined by the relation

$$b + \sum_{i=1}^n x_i w_i = 0$$

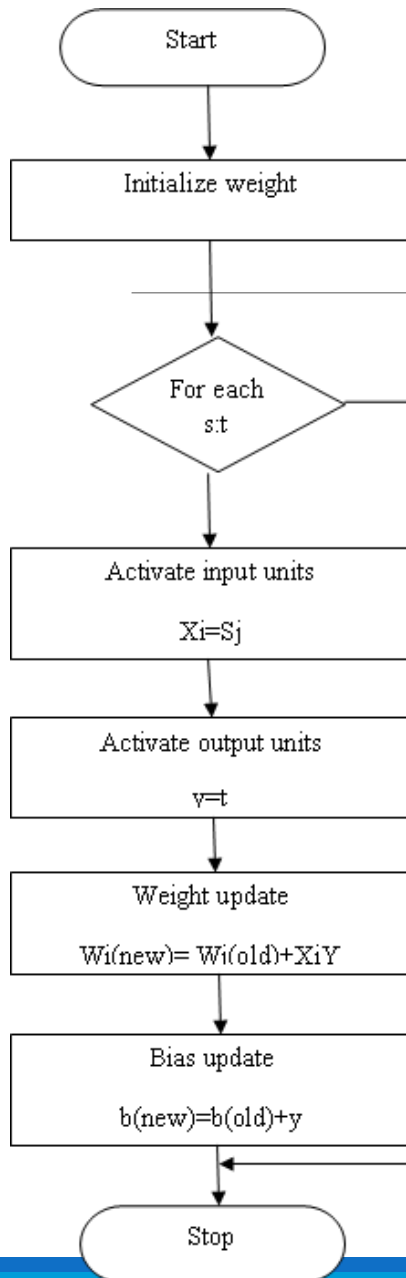
CONTD..

- Consider a network having positive response in the first quadrant and negative response in all other quadrants with either binary or bipolar data.
- Decision line is drawn separating two regions as shown in Fig.
- Using bipolar data representation, missing data can be distinguished from mistaken data. Hence bipolar data is better than binary data.
- Missing values are represented by 0 and mistakes by reversing the input values from +1 to -1 or vice versa.



HEBB NETWORK

- Donald Hebb stated in 1949 that “ *In brain, the learning is performed by the change in the synaptic gap*”.
- When an axon of cell A is near enough to excite cell B, and repeatedly or permanently takes place in firing it, some growth process or metabolic change takes place in one or both the cells such that A's efficiency, as one of the cells firing B, is increased.
- According to Hebb rule, *the weight vector is found to increase proportionately to the product of the input and the learning signal*.
- In Hebb learning, two interconnected neurons are ‘on’ simultaneously.
- The weight update in Hebb rule is given by
 - $W_i(\text{new}) = w_i(\text{old}) + x_i y$.
- It is suited more for bipolar data.
- If binary data is used, the weight updation formula cannot distinguish two conditions namely
 - A training pair in which an input unit is “on” and the target value is “off”
 - A training pair in which both the input unit and the target value is “off”.



FLOWCHART OF TRAINING ALGORITHM

Steps:

- 0: First initialize the weights.
- 1: Steps 2-4 have to be performed for each input training vector and target output pair, s:t
- 2: Input activations are set. The activation function for input layer is identity function.
 - $X_i = S_i$ for $i=1$ to n
- 3: Output activations are set.
- 4: Weight adjustment and bias adjustments are performed.
 - $W_i(\text{new}) = w_i(\text{old}) + x_i y$
 - $b(\text{new}) = b(\text{old}) + y$
- In step 4, the weight updation formula can be written in vector form as
 - $w(\text{new}) = w(\text{old}) + y.$
- Change in weight is expressed as
 - $\Delta w = xy$

Hence,

$$w(\text{new}) = w(\text{old}) + \Delta w$$

Hebb rule is used for pattern association, pattern categorization, pattern classification and over a range of other areas